

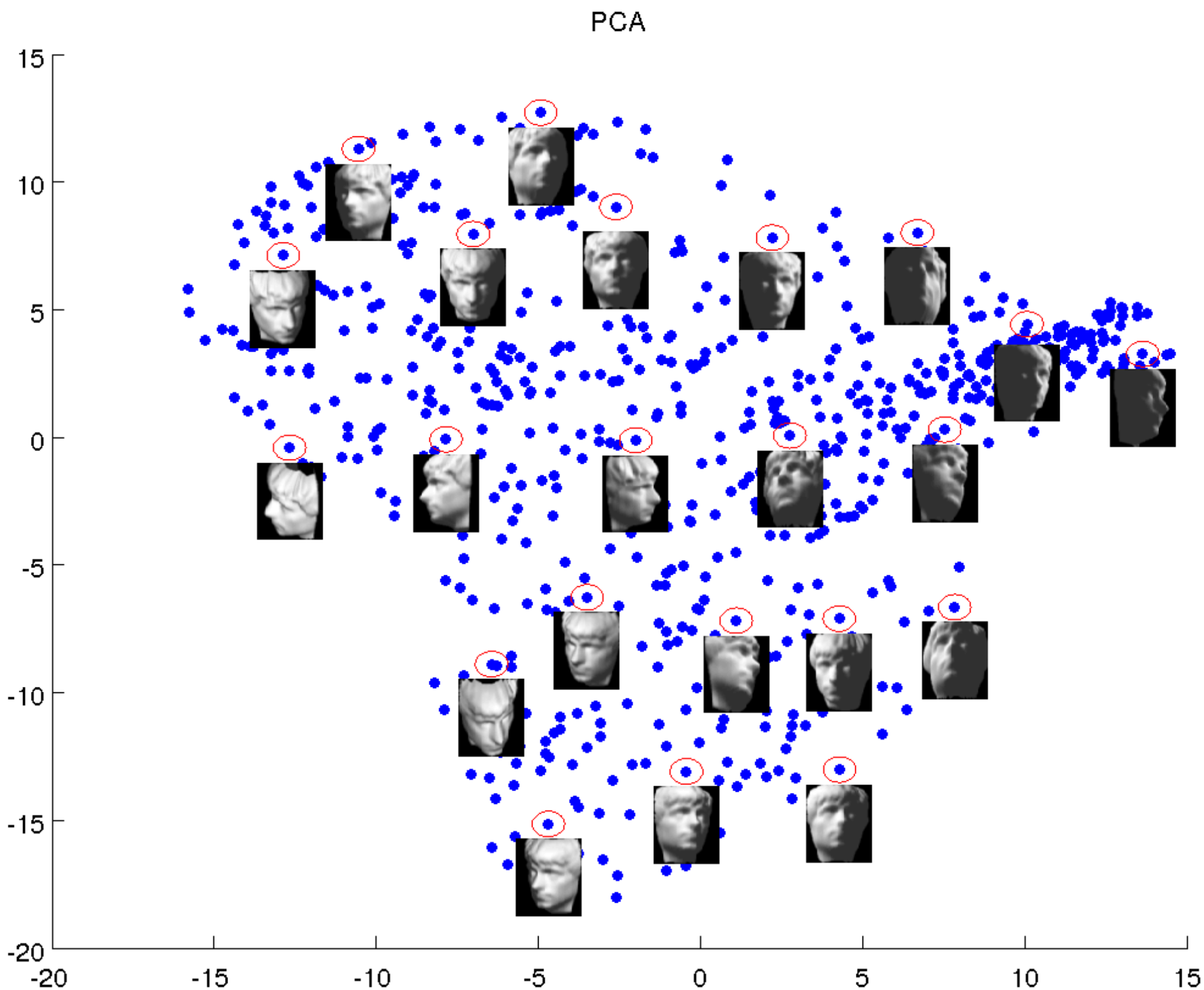
CSE 6740: Computational Data Analysis

Spring 2026

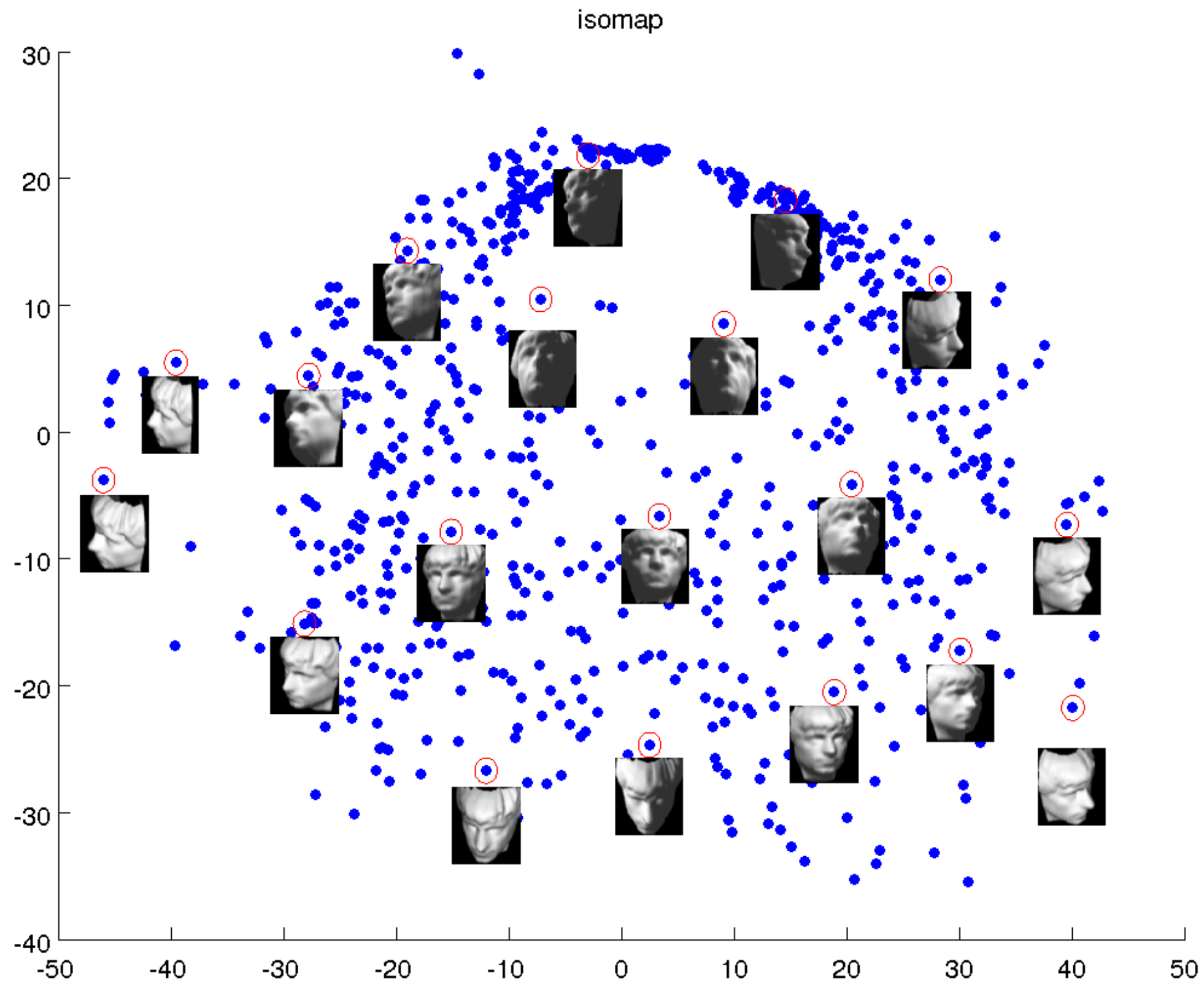
ISOMAP and Density Estimation

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02/03

Is the principal direction interpretable?

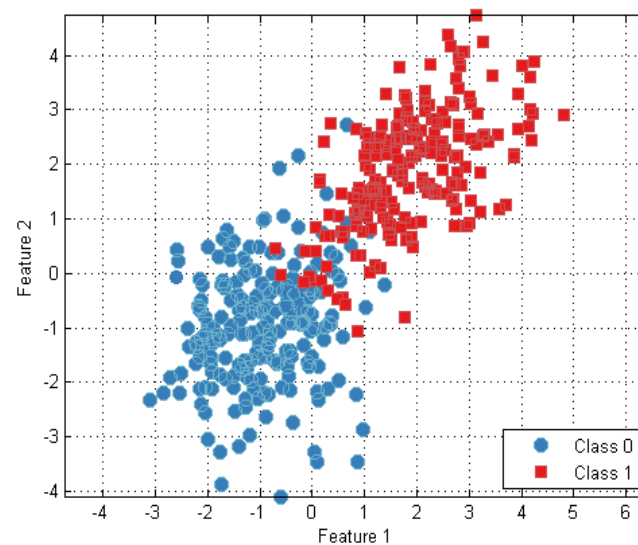


How about this one?

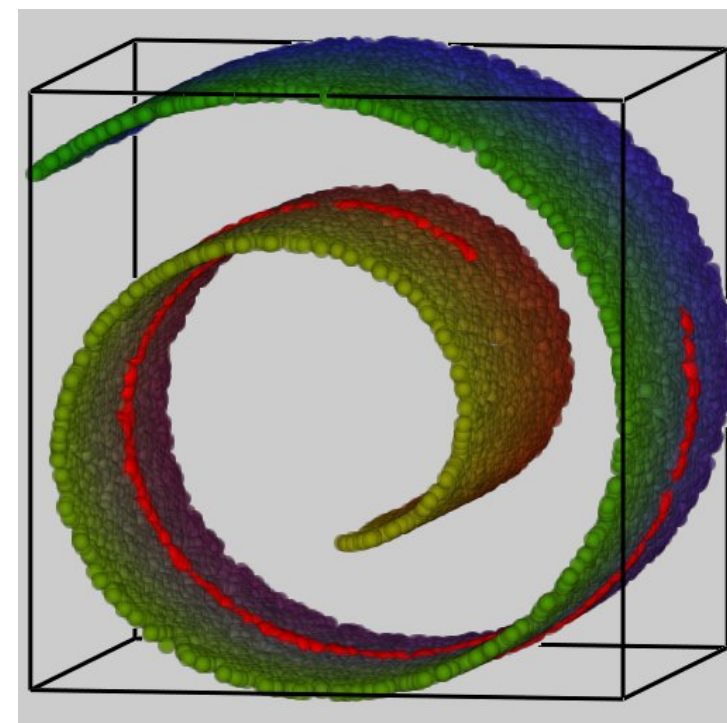
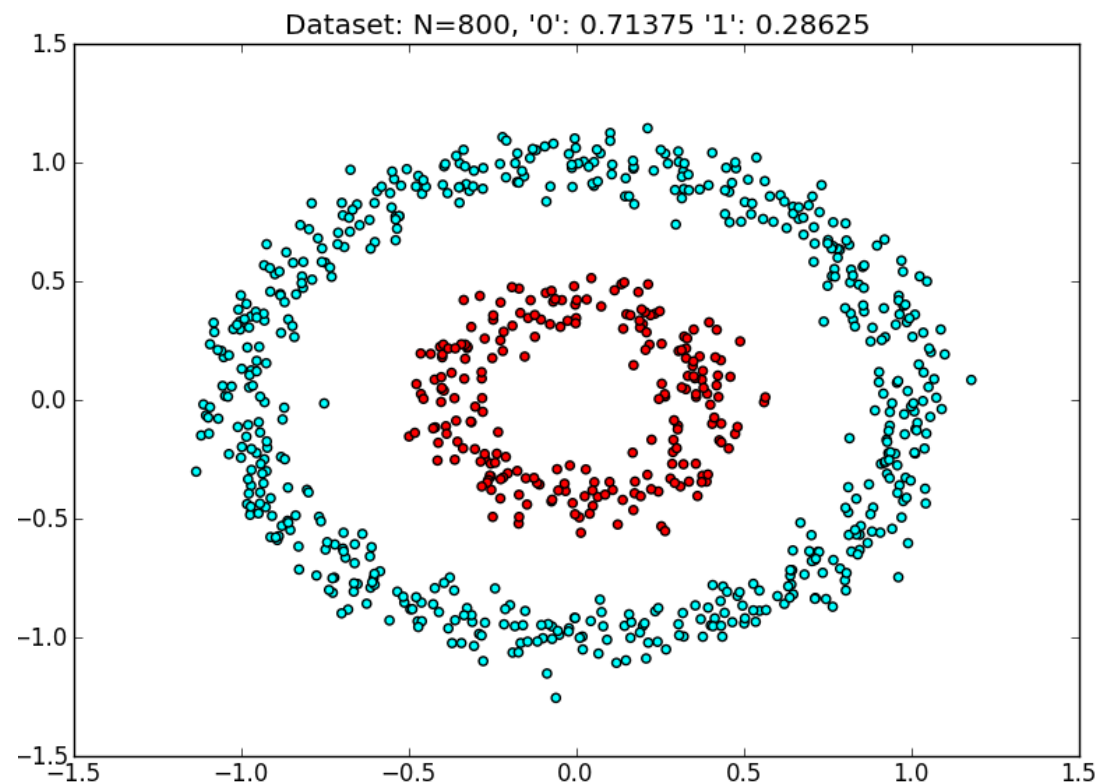


Limitation of PCA and SVD

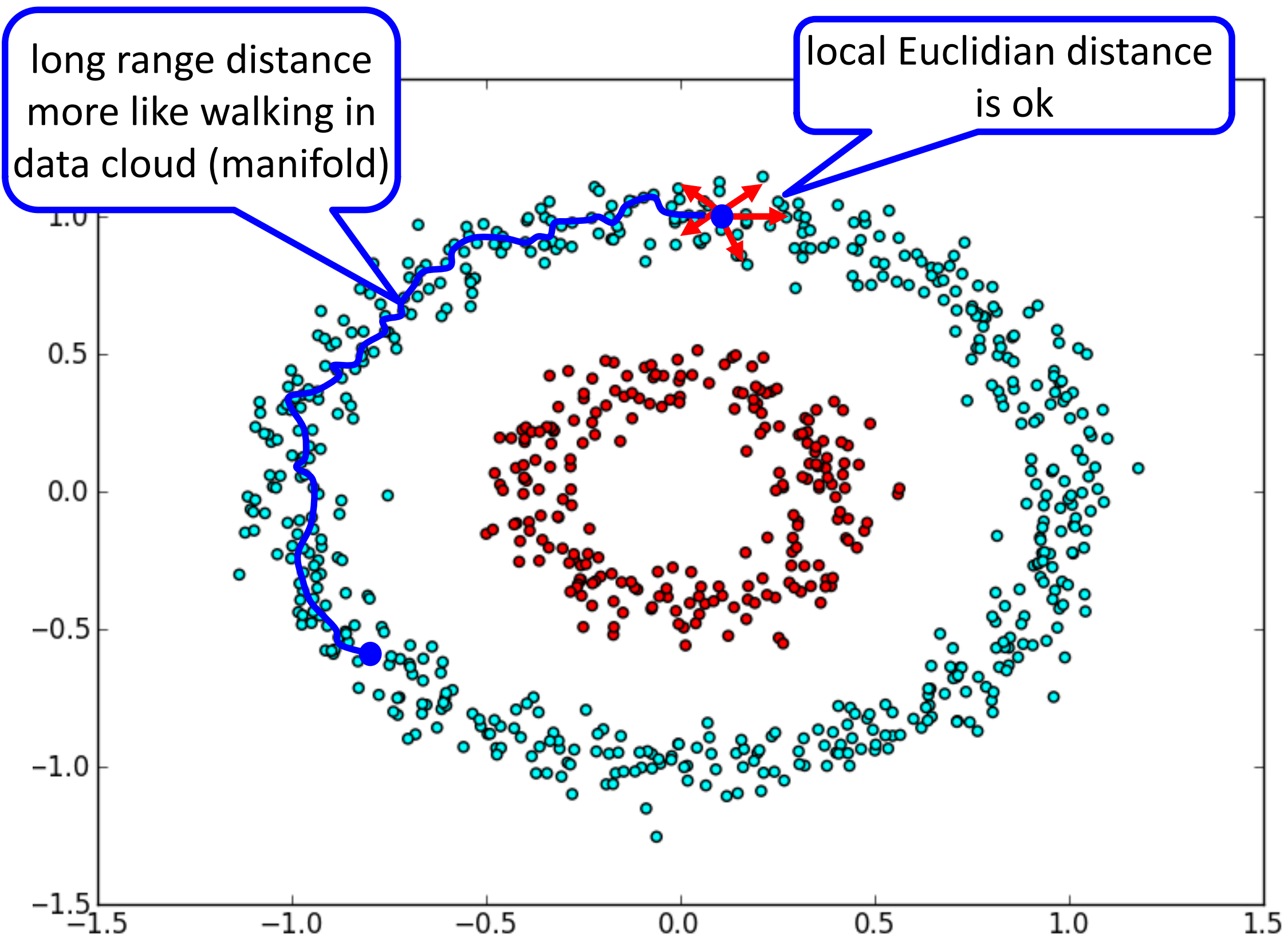
- Suitable when variables are linearly correlated



- Not suitable when nonlinear structures are present



What's a reasonable distance measure

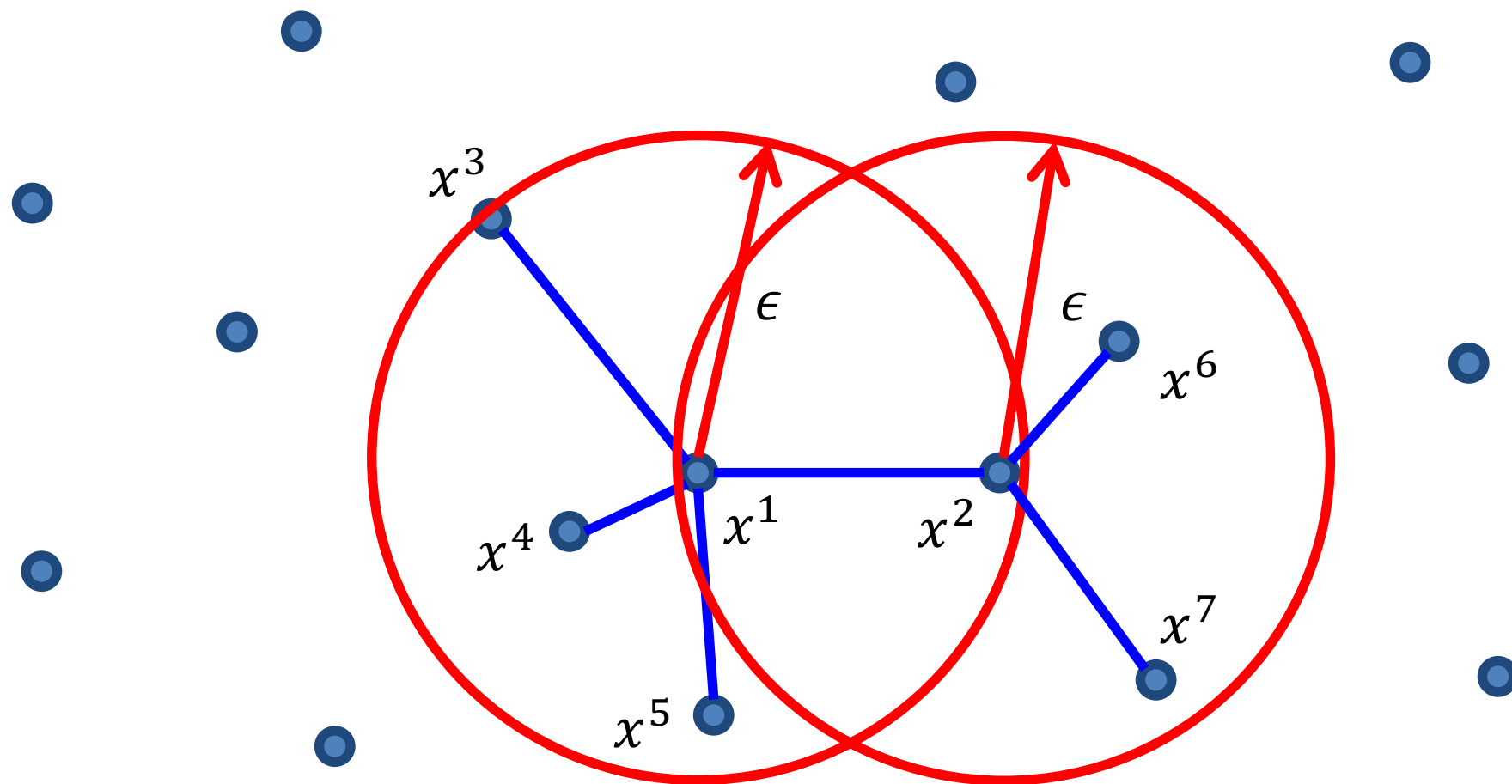


- Key idea: produce low dimensional representation which preserves “walking-distance” over the data cloud (manifold)
 - Find neighbors $N(i)$ of each data point, x^i , within distance ϵ and let A be the adjacency matrix recording neighbor **Euclidean distance**
 - Find **shortest path distance matrix** D between each pairs of points, x^i and x^j , based on A
 - Find low dimensional representation which preserves the distances information in D

Neighbor graph

- Given m data points, threshold ϵ , construct matrix $A \in R^{m \times m}$

$$A^{ij} = \begin{cases} \|x^i - x^j\|, & \text{if } \|x^i - x^j\| \leq \epsilon \\ 0, & \text{otherwise} \end{cases}$$



Shortest path distances

- With the graph defined by $A \in R^{m \times m}$, find the shortest path distance matrix D between all pairs of points
 - Also called graph distance matrix
- Can be computed with
 - Floyd-Warshall algorithm (all pair shortest path problem)
 - m applications of Dijkstra's algorithm

How to extract reduced representation?

- Now we have a distance matrix D
- At this point we can throw away the original data points and work with just D
- Can we extract from D a new set of coordinates (or reduced representation) $\tilde{Z}$ which best explains D ?
- Hints: related distance to inner product.
 - Given coordinates, we can compute Euclidean distances
 - Given Euclidean distances, can we compute the coordinates?

From coordinates to distances

- Distance to inner product

$$\begin{aligned} D^{ij^2} &= \|z^i - z^j\|^2 \\ &= (z^i - z^j)^\top (z^i - z^j) \\ &= z^{i\top} z^i + z^{j\top} z^j - 2z^{i\top} z^j \end{aligned}$$

- In matrix format, let $Z = (z^1, z^2, \dots, z^m) \in R^{n \times m}$

$$(D)^2 = a1^\top + 1a^\top - 2Z^\top Z$$

Where $a = (z^{1\top} z^1, z^{2\top} z^2, \dots, z^{m\top} z^m)^\top$

Relate distances to coordinates (or inner products)

- Construct a special centering matrix $H = I - \frac{1}{m} \mathbf{1}\mathbf{1}^\top$
 - $\left(I - \frac{1}{m} \mathbf{1}\mathbf{1}^\top\right) \mathbf{1}a^\top \left(I - \frac{1}{m} \mathbf{1}\mathbf{1}^\top\right) = 0$
 - $\left(I - \frac{1}{m} \mathbf{1}\mathbf{1}^\top\right) a\mathbf{1}^\top \left(I - \frac{1}{m} \mathbf{1}\mathbf{1}^\top\right) = 0$
- Then apply it to both side of $(D)^2$
 - $-\frac{1}{2m} H(D)^2 H = -\frac{1}{2m} H(a\mathbf{1}^\top + \mathbf{1}a^\top - 2Z^\top Z)H = \frac{1}{m} HZ^\top ZH$
 - $ZH = Z \left(I - \frac{1}{m} \mathbf{1}\mathbf{1}^\top\right) = Z - \mu \mathbf{1}^\top = \tilde{Z}$

Can we uniquely recover the coordinates? ✓

- Given $C = -\frac{1}{2}H(D)^2H = \tilde{Z}^T \tilde{Z}$ $X^T Z^T Z X \geq 0$
 $= V^T V \geq 0 = \sum_{i=1}^n v_i^2 \geq 0$
- Multiple coordinate matrices $\tilde{Z}$ result in the same C
- For any orthogonal matrix B :

$$C = \tilde{Z}^T \tilde{Z} = \tilde{Z}^T B^T B \tilde{Z}$$
- An orthogonal matrix is like a transformation of the data
- Therefore, we can hope to recover the coordinate matrix $\tilde{Z}$ up to some transformation

Obtain low dimensional representation

- Given $C = -\frac{1}{2}H(D)^2H = \tilde{Z}^T \tilde{Z}$

$$= \sum_{i=1}^d z_i z_i^T = \sum_{i=1}^d \lambda_1 \underline{w_1 w_1^T} + \lambda_2 \underline{w_2 w_2^T} + \dots$$
- Pick a **particular** coordinate matrix $\tilde{Z}$ that best approximates C

- Perform eigenvalue decomposition of C

- $Cw = \lambda w$
- Take the eigenvectors $w^1, w^2, \dots$ of C corresponding to
 - the largest eigenvalue λ_1 , as the first coordinate
 - the second largest eigenvalue λ_2 , as the second coordinate

- Reduced representation

$$\tilde{Z}^T = (w^1, w^2, \dots) \begin{pmatrix} \lambda_1^{1/2} & & \\ & \lambda_2^{1/2} & \\ & & \ddots \end{pmatrix} = \begin{pmatrix} \lambda_1^{1/2} w_1 & \lambda_2^{1/2} w_2 & \dots \end{pmatrix}$$

Squared Euclidean Distances Yield a PSD Matrix

Let $D^2 \in \mathbb{R}^{m \times m}$ be a squared distance matrix with

$$D_{ij}^2 = \|z_i - z_j\|_2^2,$$

and define the centering matrix

$$H = I - \frac{1}{m} \mathbf{1}\mathbf{1}^\top.$$

Key result:

$$C := -\frac{1}{2} H D^2 H \succeq 0.$$

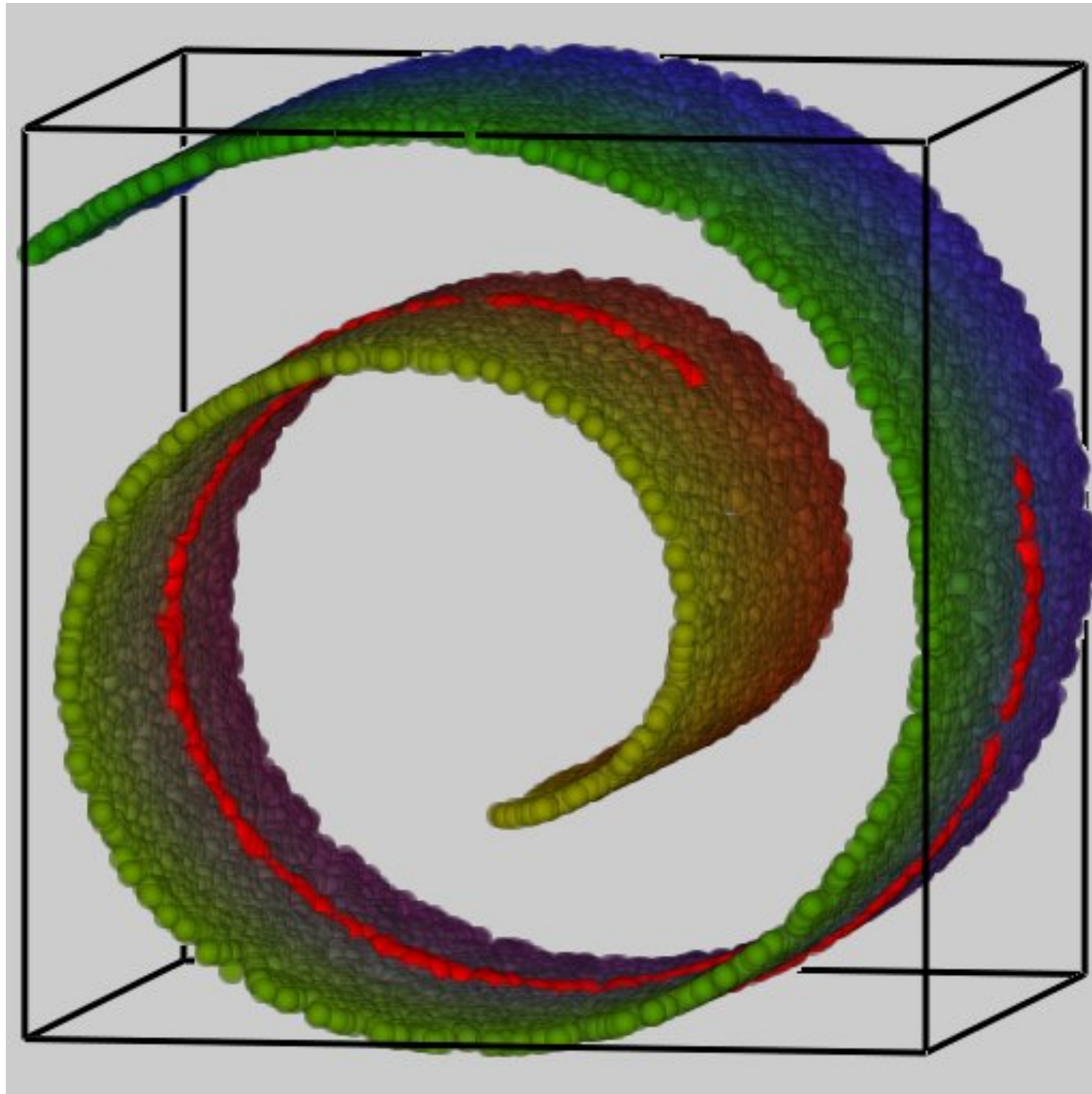
Implication:

- C is positive semidefinite
- All eigenvalues of C are nonnegative
- Coordinates can be recovered via eigendecomposition

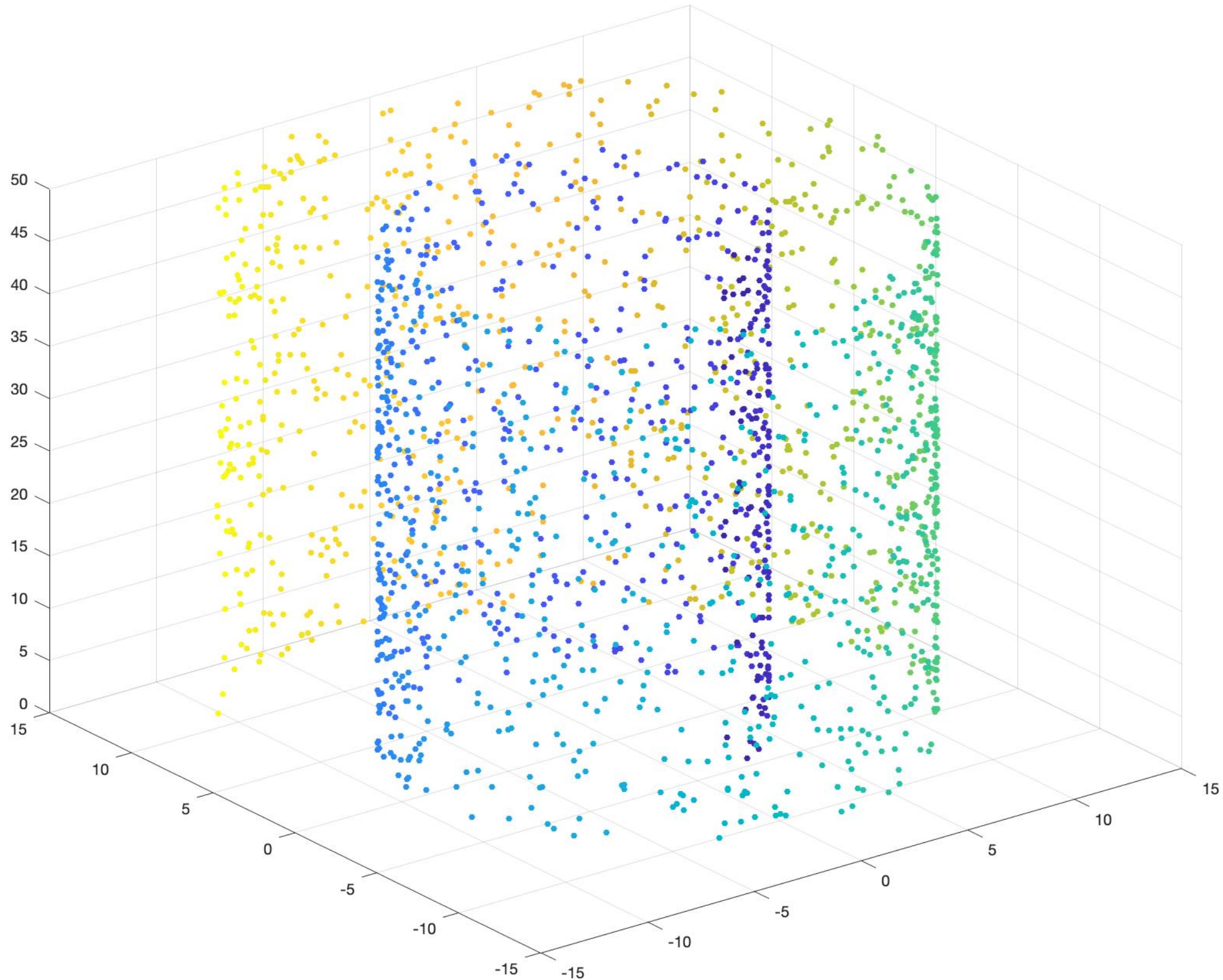
- Given m data points, $\{x^1, x^2, \dots x^m\} \in R^n$
- Step 1: build a **weighted** graph A using nearest neighbors, and compute pairwise shortest distance matrix D
- Step 3: use a centering matrix $H = I - \frac{1}{m} \mathbf{1}\mathbf{1}^\top$ to get
$$C = -\frac{1}{2m} H(D)^2 H$$
- Step 4: compute leading eigenvectors $w^1, w^2, \dots$ and eigenvalues $\lambda_1, \lambda_2, \dots$ of C

$$Z^\top = (w^1, w^2, \dots) \begin{pmatrix} \lambda_1^{1/2} & & \\ & \lambda_2^{1/2} & \\ & & \ddots \end{pmatrix}$$

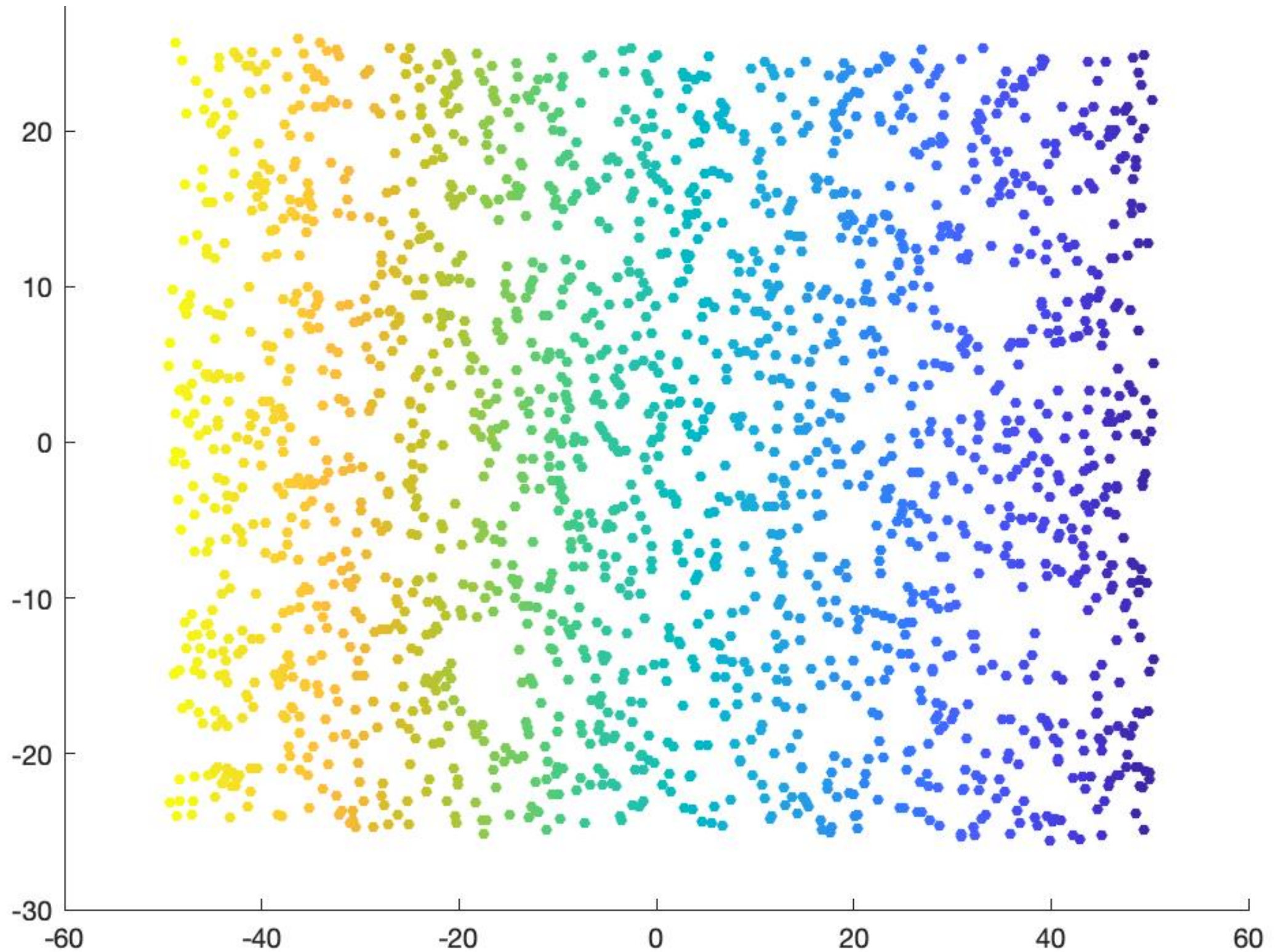
Swissroll(demo test_isomap2.m)



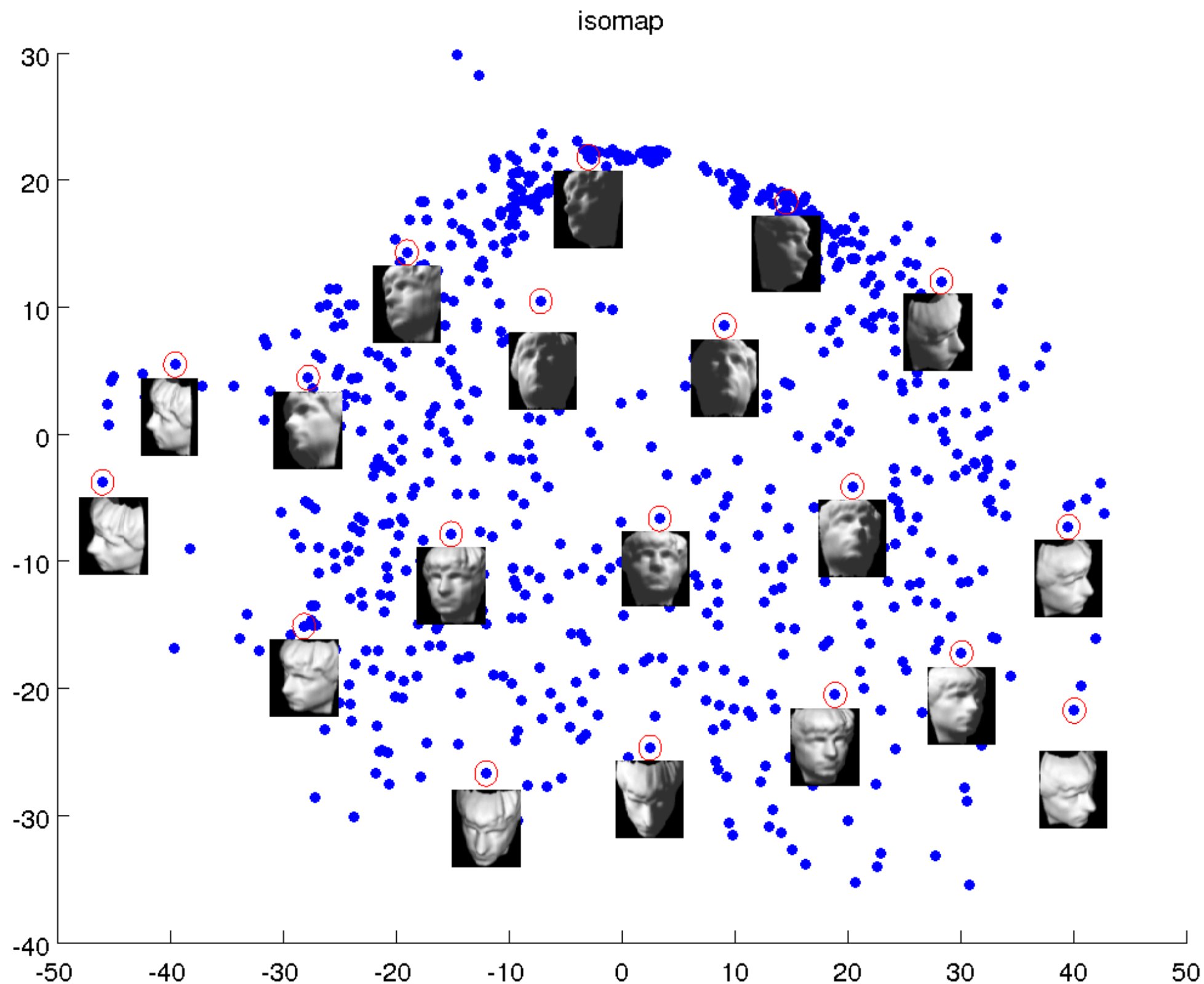
Swissroll(demo test_isomap2.m)



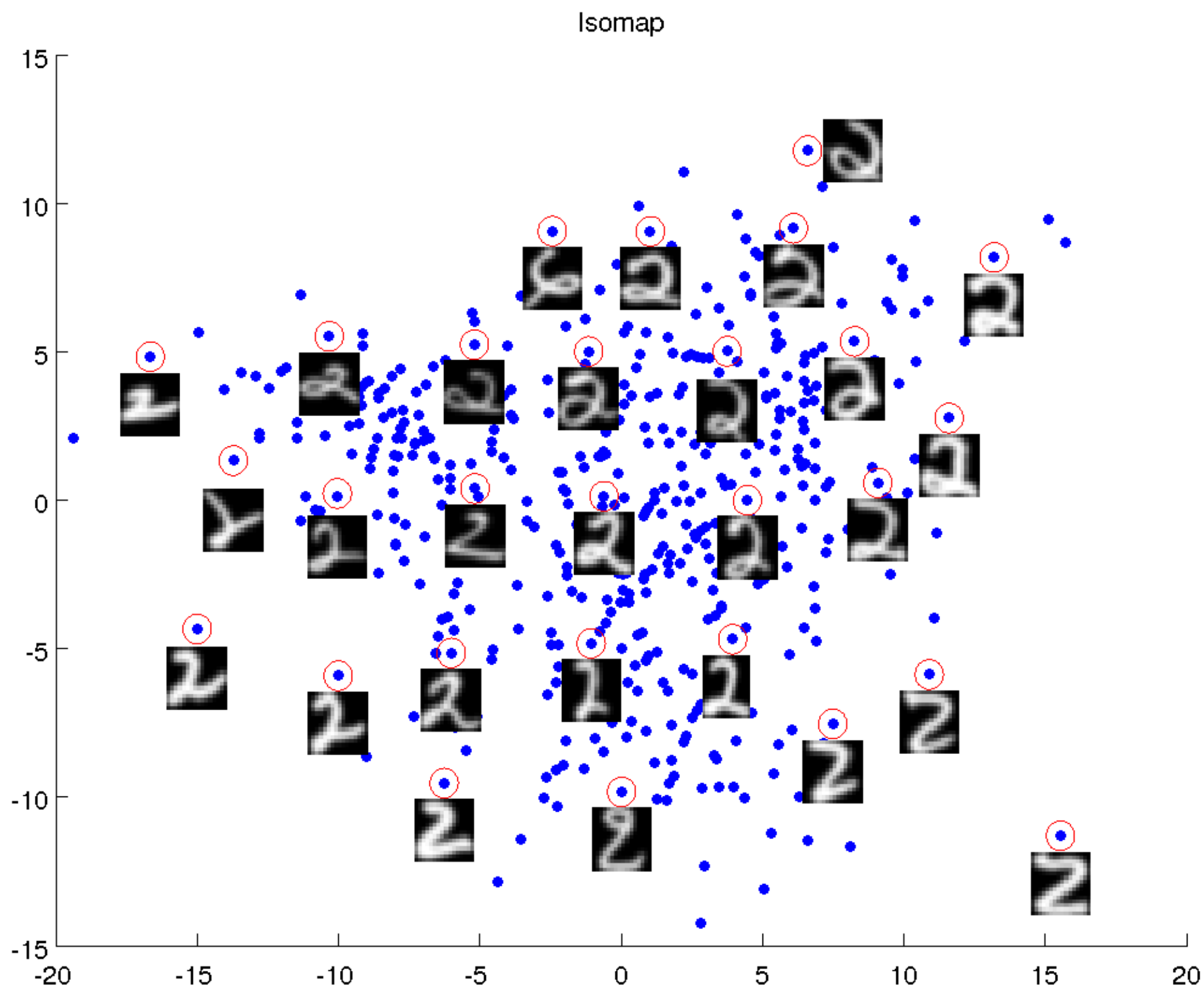
Swissroll(demo test_isomap2.m)



Faces (demo face_isomap_pca.m)

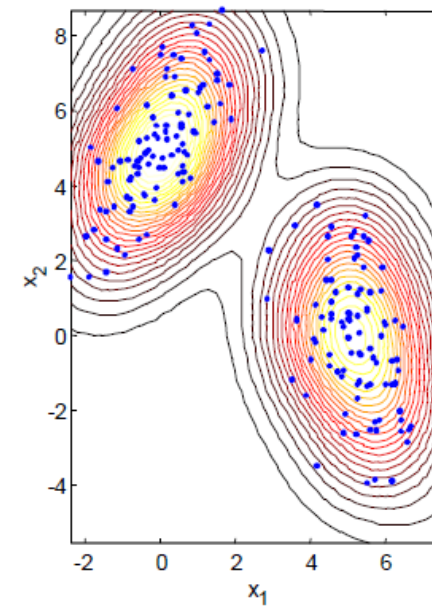
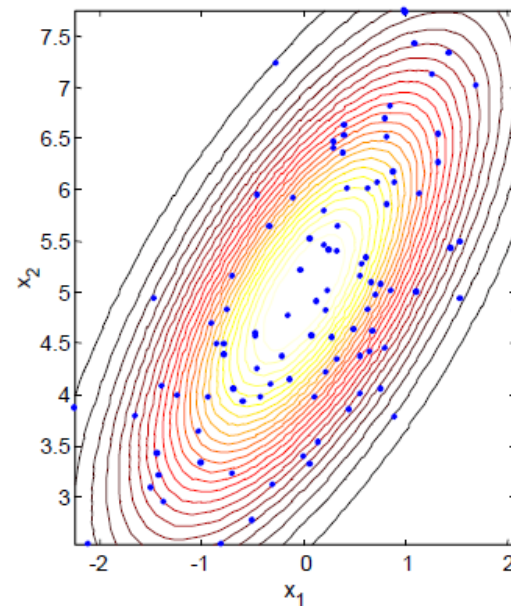


Handwritten digits (demo usps_isomap_pca.m)



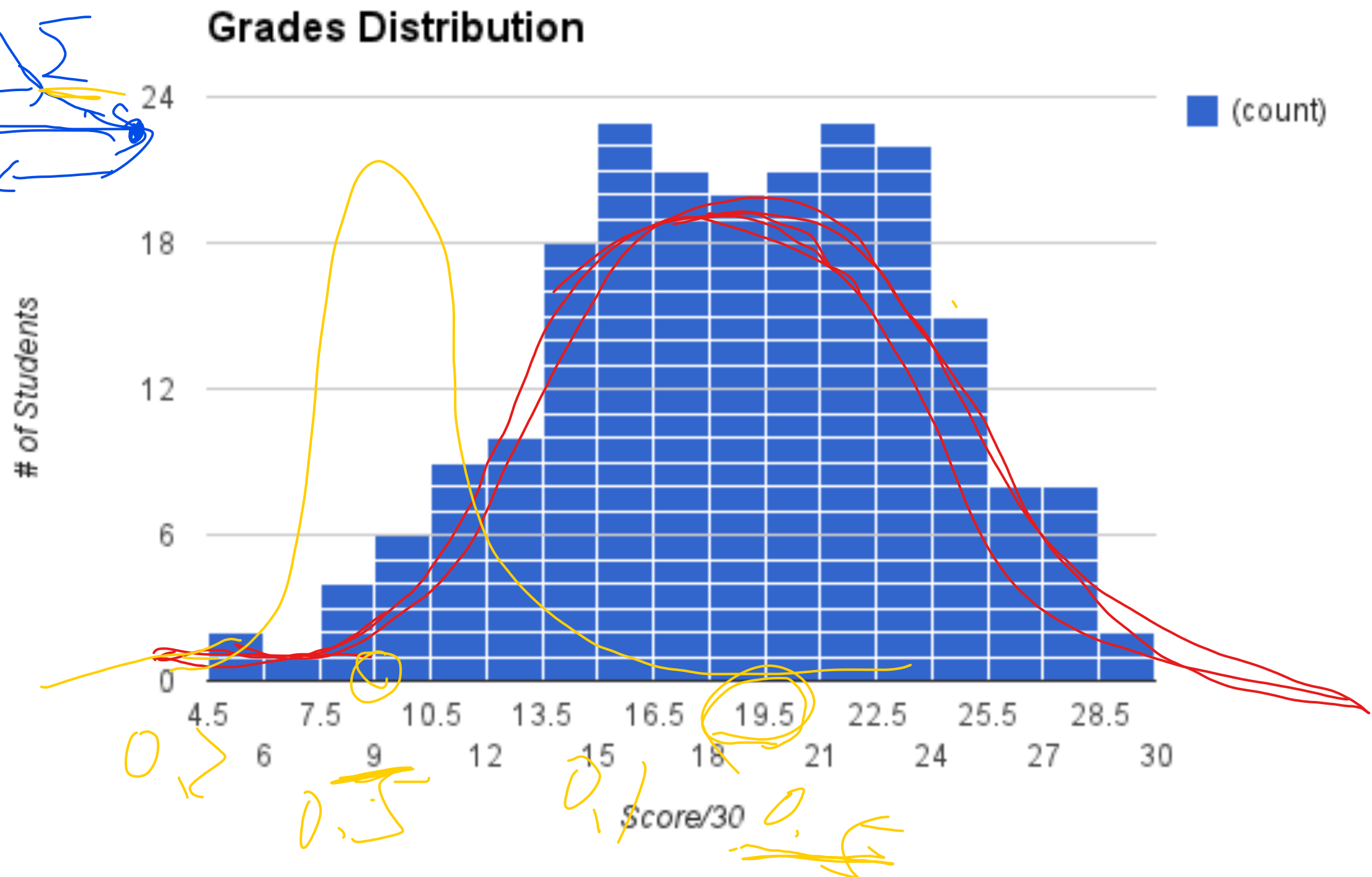
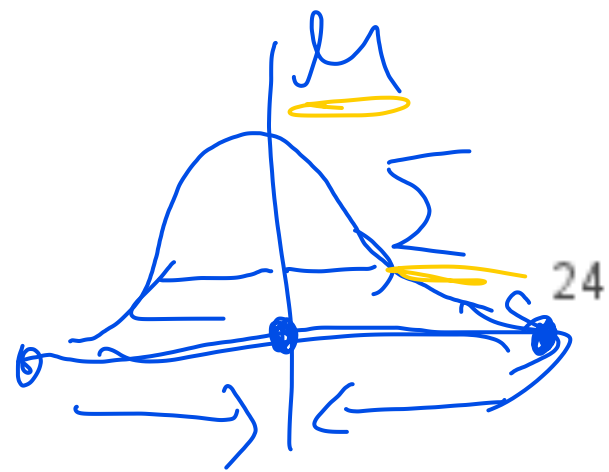
Why do we need density estimation?

- Learn more about the “shape” of the data cloud

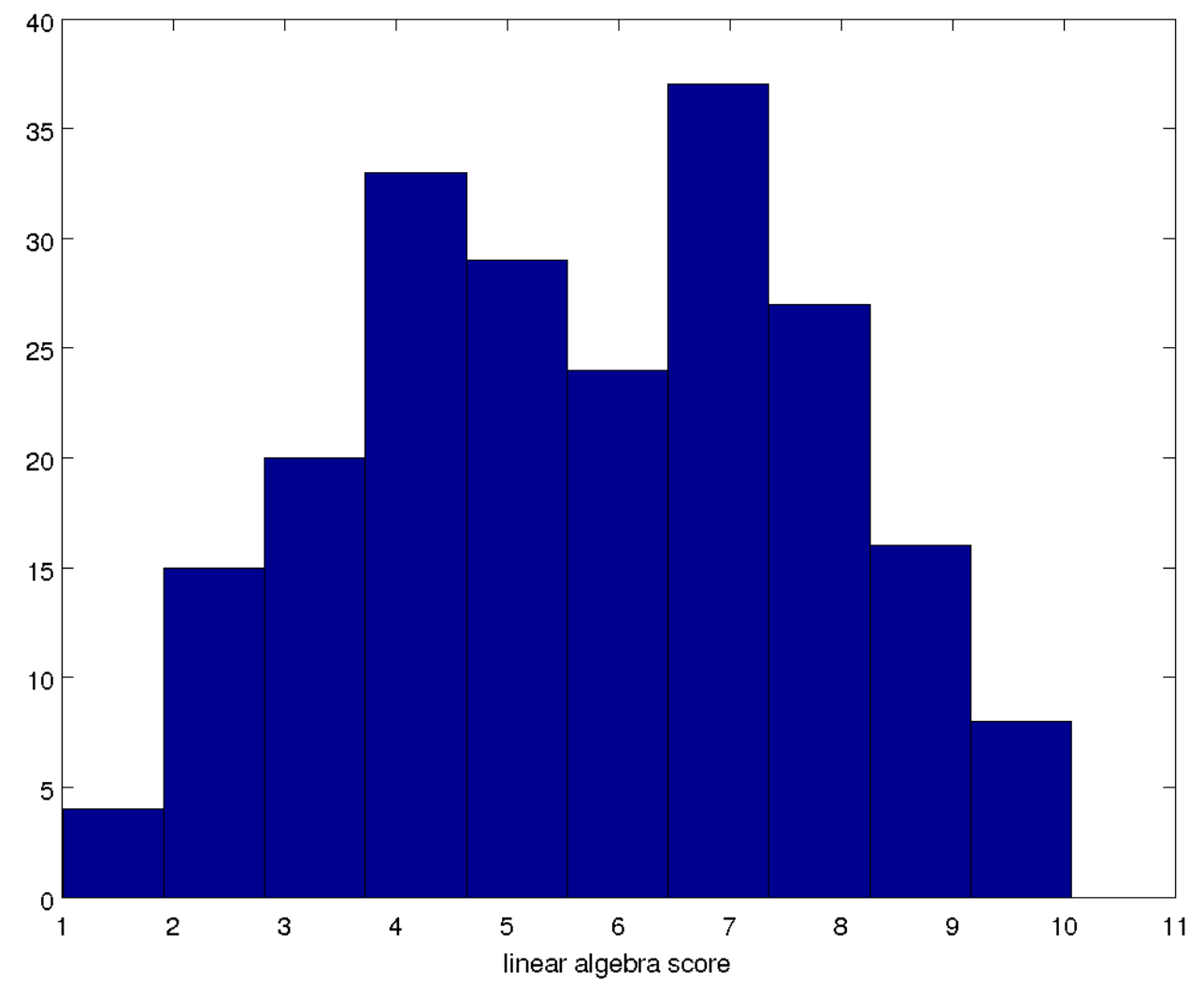
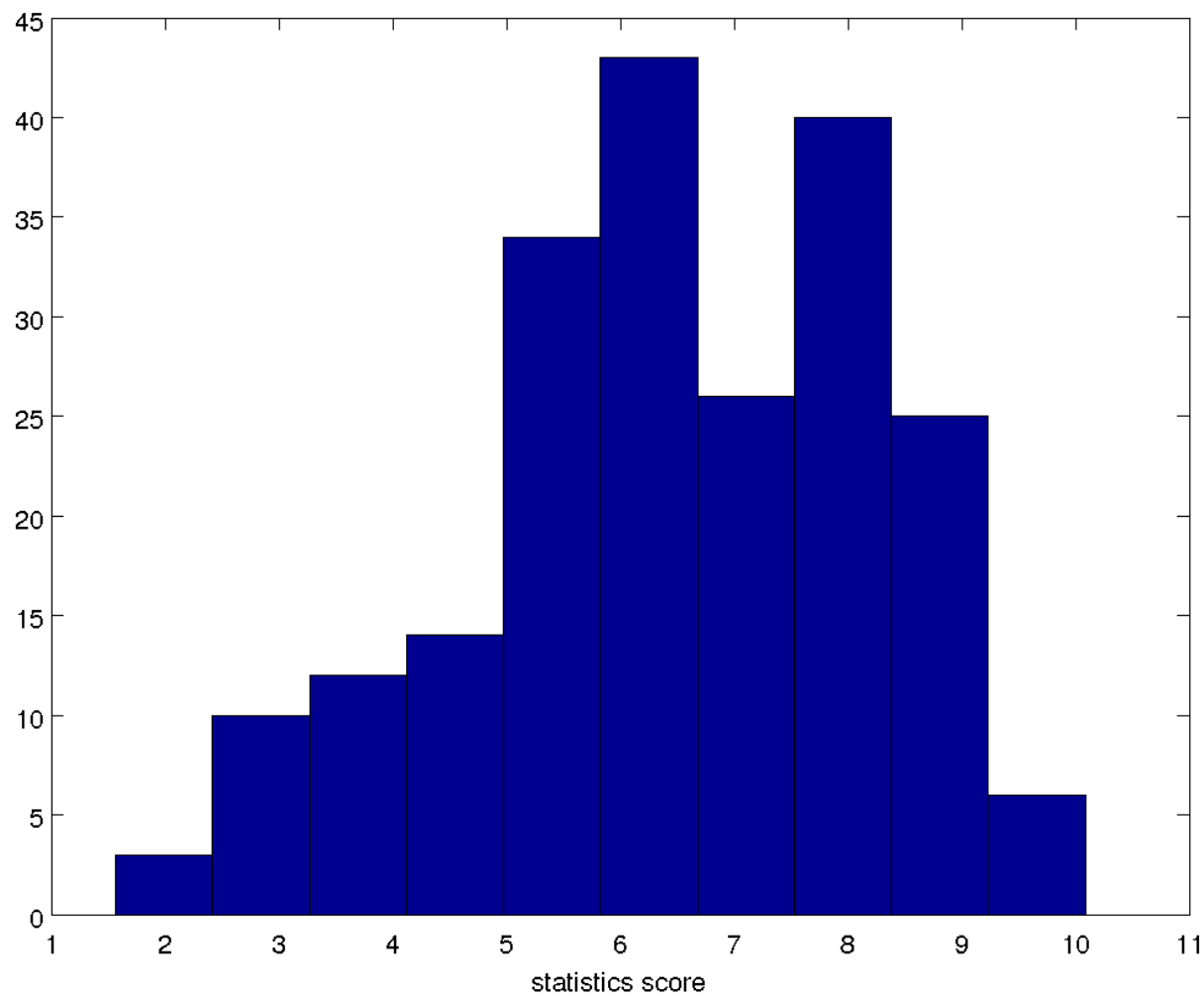


- Assess the likelihood of seeing a particular data point
 - Is this a typical data point? (high density value)
 - Is this an abnormal data point / outlier? (low density value)
- Building block for more sophisticated learning algorithms
 - Classification, regression, graphical models ...
 - A simple recommendation system

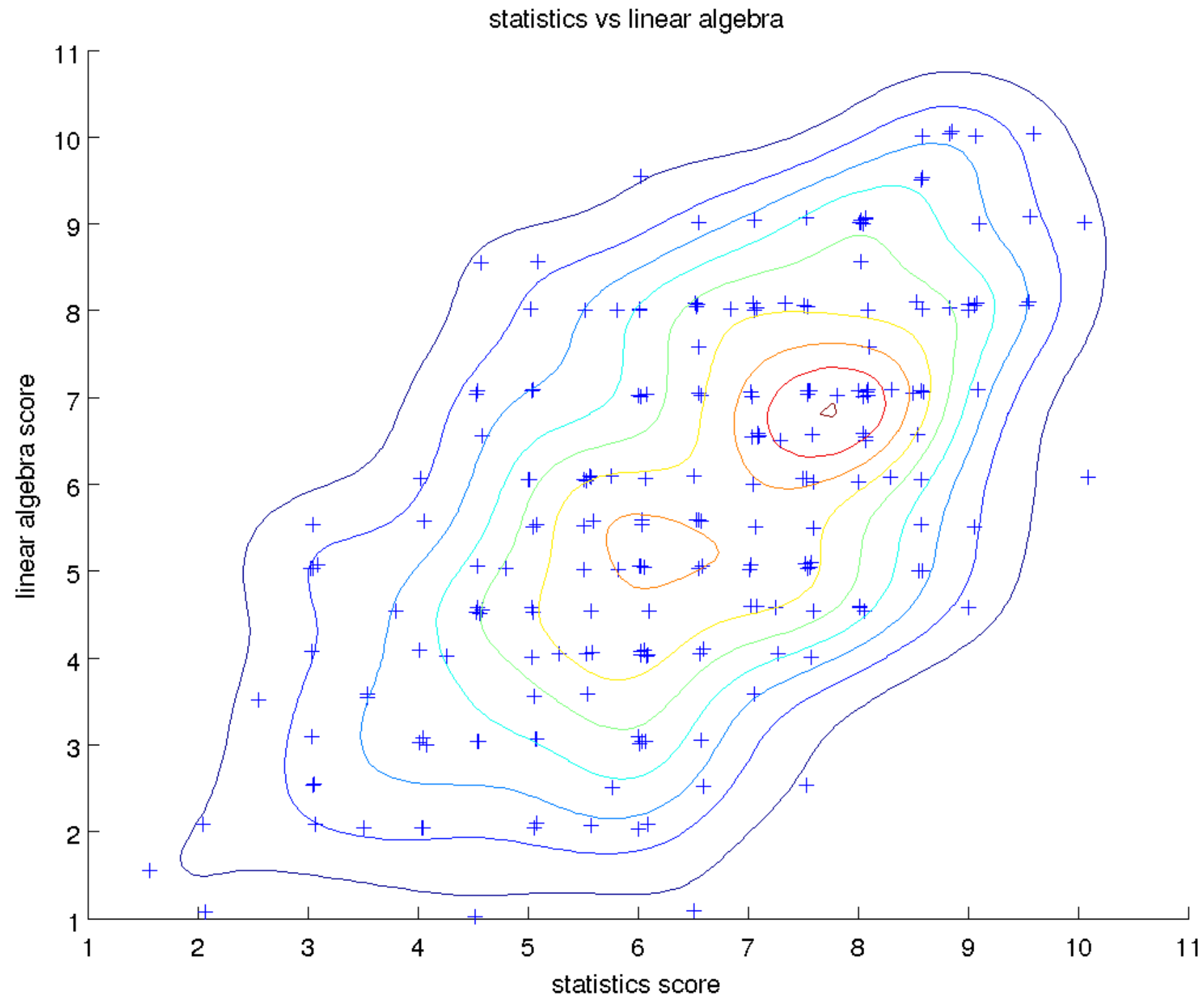
Example: test scores



Example: test scores (cont.)



Example: test scores (cont.)



Parametric models

- Models which can be described by a fixed number of parameters

- Discrete case: eg. Bernoulli distribution

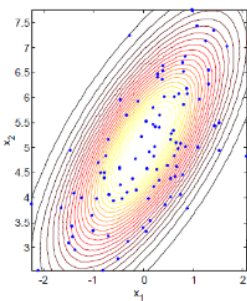
$$P(x|\theta) = \theta^x (1 - \theta)^{1-x}$$

one parameter, $\theta \in [0,1]$, which generate a family of models, $\mathcal{F} = \{P(x|\theta) \mid \theta \in [0,1]\}$,



- Continuous case: eg. Gaussian distribution in R^n

$$p(x|\mu, \Sigma) = \frac{1}{|\Sigma|^{\frac{1}{2}} (2\pi)^{\frac{n}{2}}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$



Two sets of parameters $\{\mu, \Sigma\}$, which again generate a family of models, $\mathcal{F} = \{p(x|\mu, \Sigma) \mid \mu \in R^n, \Sigma \in R^{n \times n} \text{ and } PSD\}$,

Estimation of parametric models

- A very popular estimator is the **maximum likelihood estimator (MLE)**, which is simple and has good statistical properties
- Assume that m data points $\mathcal{D} = \{x^1, x^2, \dots, x^m\}$ drawn **independently and identically (iid)** from some distribution $P^*(x)$
- Want to fit the data with a model $P(x|\theta)$ with parameter θ

$$\theta = \underset{\theta}{\operatorname{argmax}} \log P(\mathcal{D}|\theta) = \underset{\theta}{\operatorname{argmax}} \log \prod_{i=1}^m P(x^i|\theta)$$

~~$P(x^2|x^1)P(x^1) \neq P(x^1, x^2) \neq P(x^1)P(x^2)$~~

Example problem

- Estimate the probability θ of landing in heads using a biased coin
- Given a sequence of m independently and identically distributed (iid) flips
 - Eg., $\mathcal{D} = \{x^1, x^2, \dots, x^m\} = \{1, 0, 1, \dots, 0\}, x^i \in \{0, 1\}$
- Model: $P(x|\theta) = \theta^x(1 - \theta)^{1-x}$
 - $P(x|\theta) = \begin{cases} 1 - \theta, & \text{for } x = 0 \\ \theta, & \text{for } x = 1 \end{cases}$
- Likelihood of a single observation x^i ?
 - $P(x^i|\theta) = \theta^{x^i}(1 - \theta)^{1-x^i}$



MLE for Biased Coin

- Objective function, log likelihood

$$\begin{aligned}
 l(\theta; \mathcal{D}) &= \log P(\mathcal{D}|\theta) = \log \theta^{n_h} (1-\theta)^{n_t} \\
 &= \log \prod_{i=1}^m \theta^{x^i} (1-\theta)^{1-x^i} = \log \prod_{i \in H} \theta^{x^i} \prod_{i \in T} (1-\theta)^{1-x^i} \\
 &= \log \theta^{\sum_{i \in H} x^i} (1-\theta)^{\sum_{i \in T} 1-x^i}
 \end{aligned}$$

n_h : number of heads, n_t : number of tails

- Maximize $l(\theta; \mathcal{D})$ w.r.t. θ
- Take derivatives w.r.t. θ

$$\frac{\partial l}{\partial \theta} = \frac{n_h}{\theta} - \frac{(m - n_h)}{1 - \theta} = 0$$

$$\Rightarrow \hat{\theta}_{MLE} = \frac{n_h}{m} \text{ or } \hat{\theta}_{MLE} = \frac{1}{m} \sum_i x^i$$

Estimating Gaussian distribution

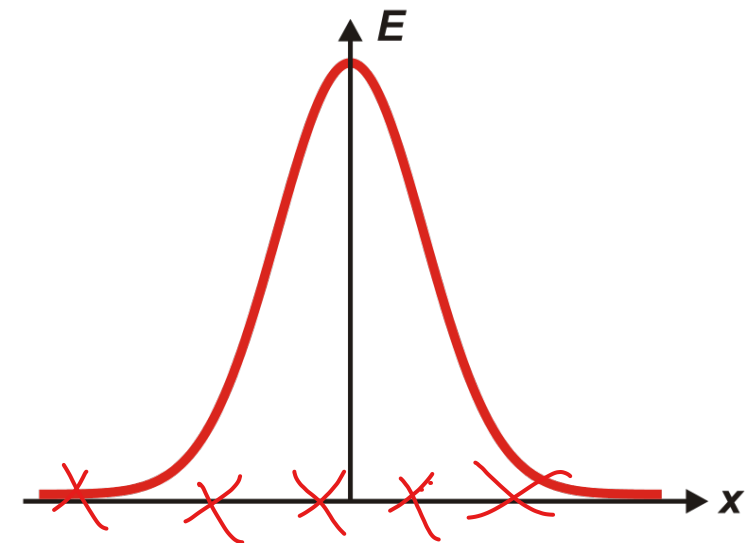
- Gaussian distribution in R

$$p(x|\mu, \sigma) = \frac{1}{(2\pi)^{\frac{1}{2}}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right)$$

- Need to estimate two sets of parameters μ, σ

- Given m iid samples

$$\mathcal{D} = \{x^1, x^2, \dots, x^m\}, x^i \in R$$



- Likelihood of one data point:

$$p(x^i|\mu, \sigma) \propto \exp\left(-\frac{1}{2\sigma^2}(x^i - \mu)^2\right)$$

The estimators for μ, σ are well-known

- Gaussian distribution in R

$$p(x|\mu, \sigma) = \frac{1}{(2\pi)^{\frac{1}{2}}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right)$$

- Mean

$$\mu = \frac{1}{m} \sum_{i=1}^m x^i$$

- Variance

$$\sigma^2 = \frac{1}{m} \sum_{i=1}^m (x^i - \mu)^2$$

MLE for Gaussian distribution

- Objective function, log likelihood

$$\begin{aligned} l(\mu, \sigma; \mathcal{D}) &= \log \prod_{i=1}^m \frac{1}{(2\pi)^{\frac{1}{2}} \sigma} \exp \left(-\frac{1}{2\sigma^2} (x^i - \mu)^2 \right) \\ &= -\frac{m}{2} \log 2\pi - \frac{m}{2} \log \sigma^2 - \sum_{i=1}^m \frac{(x^i - \mu)^2}{2\sigma^2} \end{aligned}$$

- Maximize $l(\mu, \sigma; \mathcal{D})$ with respect to μ, σ
- Take derivatives w.r.t. μ, σ^2

$$\begin{aligned} \frac{\partial l}{\partial \mu} &= 0 \\ \frac{\partial l}{\partial \sigma^2} &= 0 \end{aligned}$$

$$l(\mu, \sigma; \mathcal{D}) = -\frac{m}{2} \log 2\pi - \frac{m}{2} \log \sigma^2 - \sum_{i=1}^m \frac{(x^i - \mu)^2}{2\sigma^2}$$

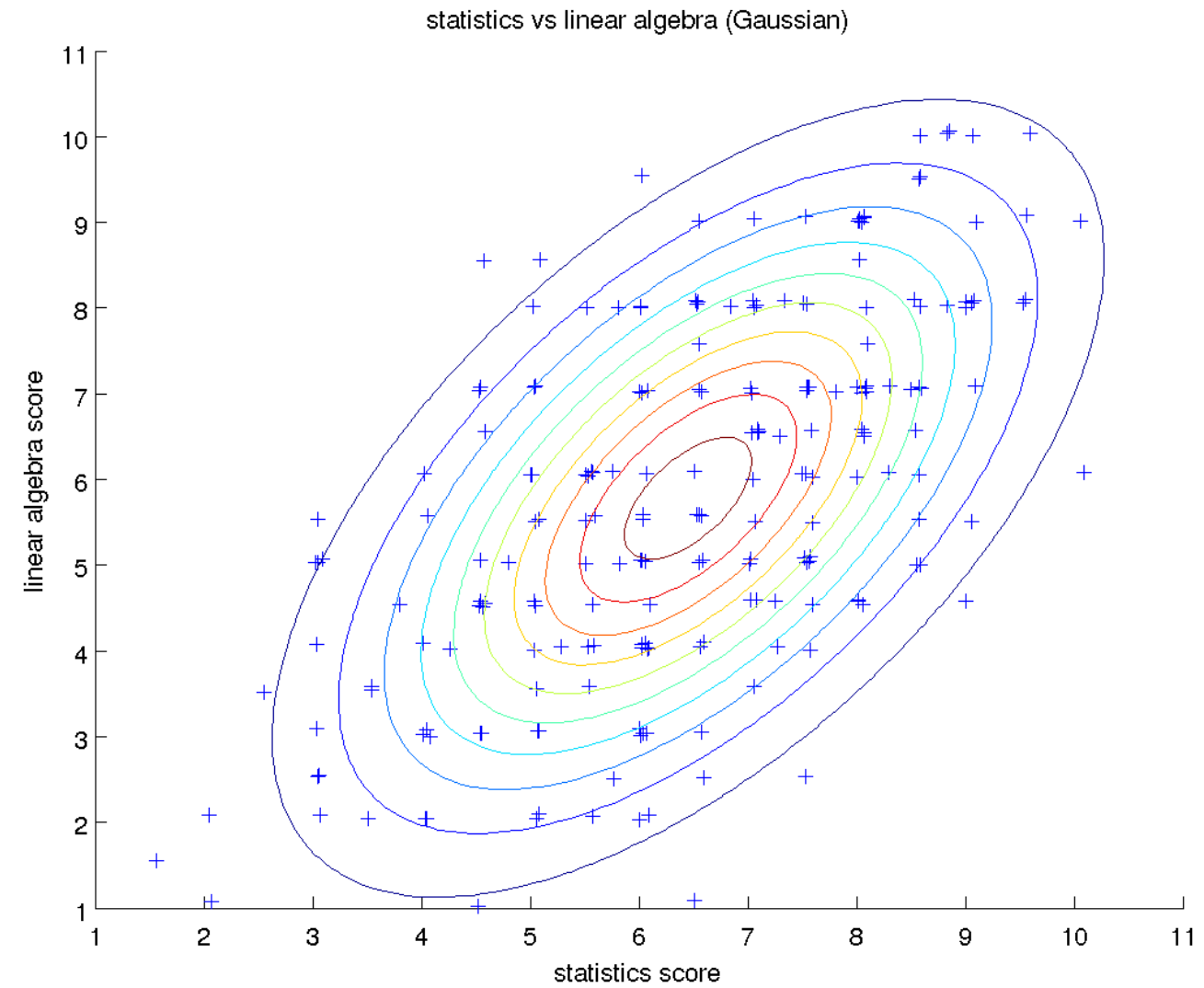
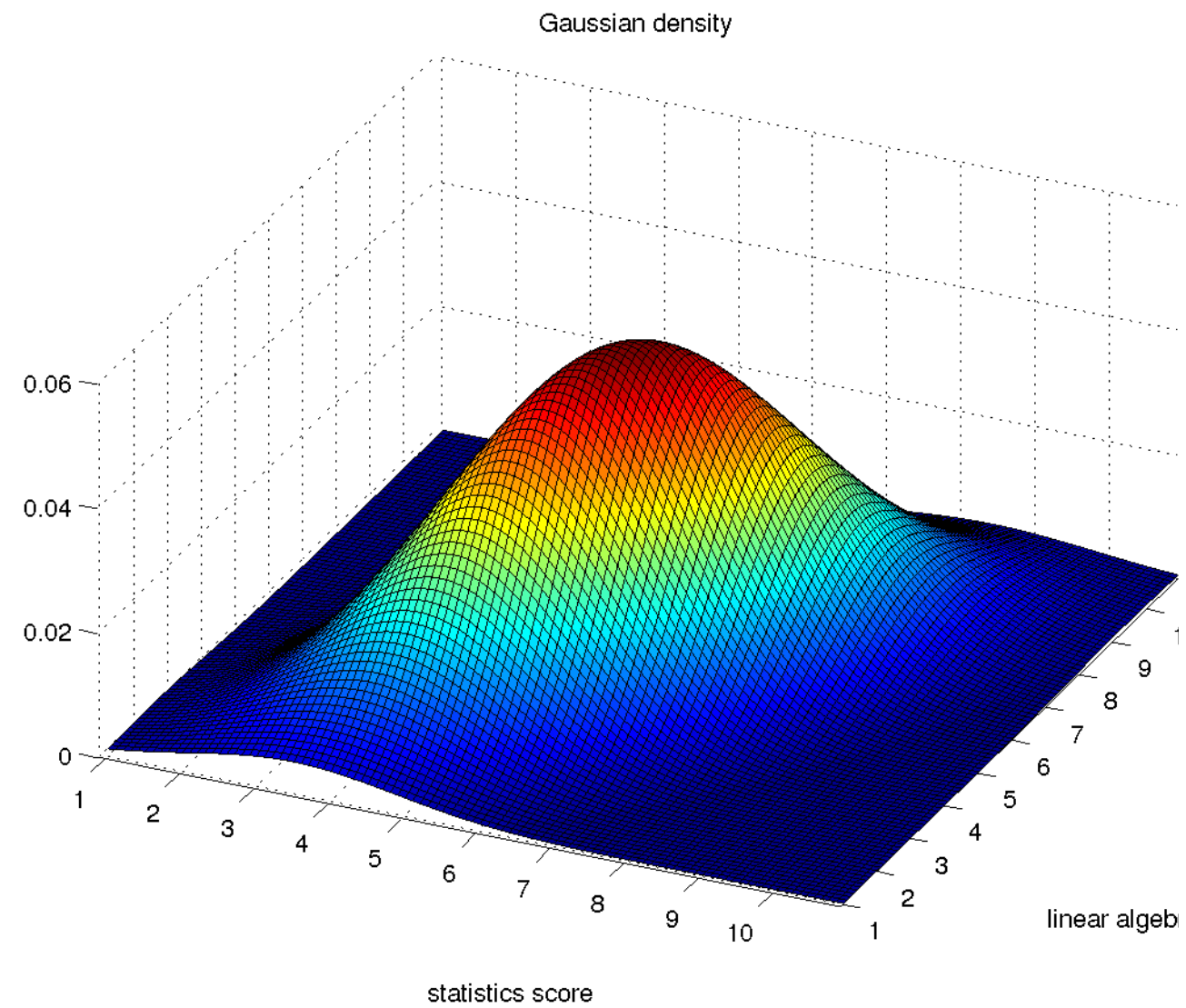
$$\frac{\partial l}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^m (x^i - \mu) = 0$$

$$\Rightarrow \sum_i x^i = m\mu \Rightarrow \mu = \frac{1}{m} \sum_{i=1}^m x^i$$

$$\frac{\partial l}{\partial \sigma^2} = -\frac{m}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_i (x^i - \mu)^2 = 0$$

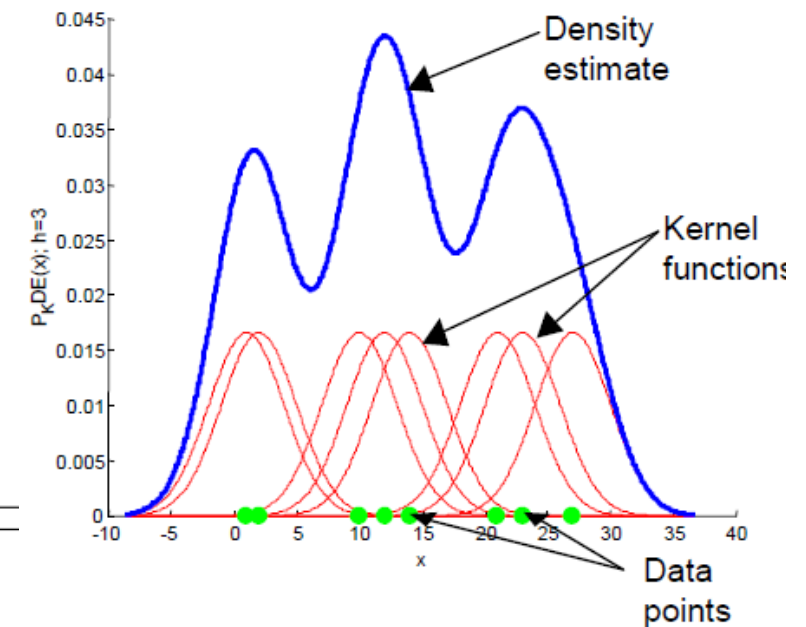
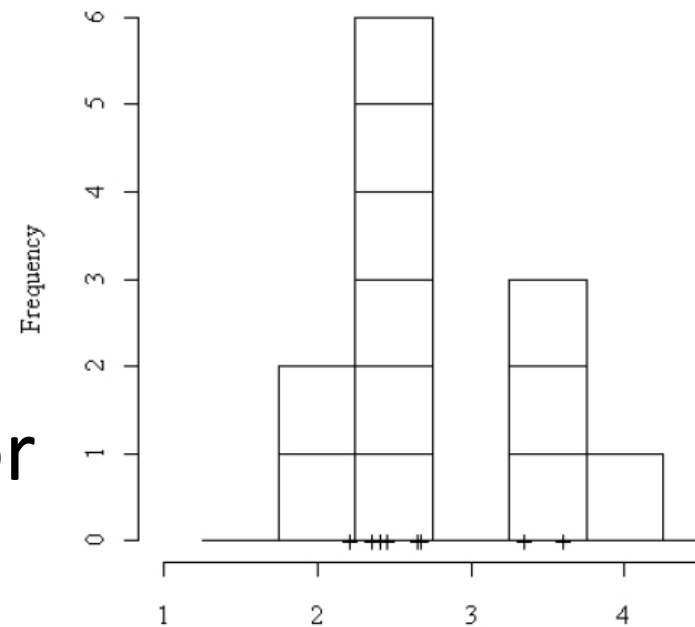
$$\Rightarrow \sum_i (x^i - \mu)^2 = m\sigma^2 \Rightarrow \frac{1}{m} \sum_{i=1}^m (x^i - \mu)^2$$

Density example



Nonparametric models

- Eg. Histogram
- Eg. Kernel density estimator



- What are nonparametric models?
 - “nonparametric” does **not** mean there are no parameters
 - can not be described by a fixed number of parameters
 - one can think of there are many many parameters

1-D Histogram

- One the simplest nonparametric density estimator

- Given m iid samples $\mathcal{D} = \{x^1, x^2, \dots, x^m\}, x^i \in [0,1)$

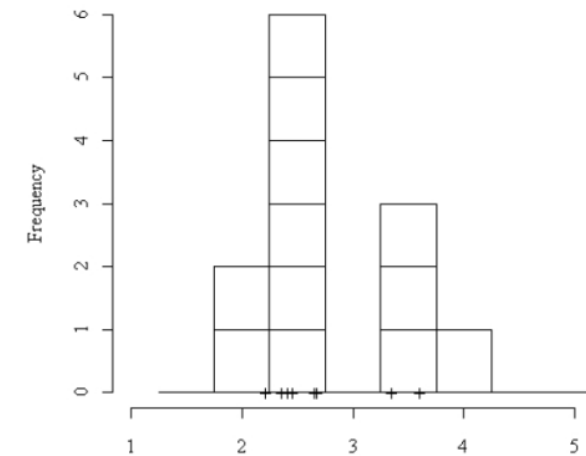
- Split $[0,1)$ into n bins

$$B_1 = \left[0, \frac{1}{n}\right), B_2 = \left[\frac{1}{n}, \frac{2}{n}\right), \dots, B_n = \left[\frac{n-1}{n}, 1\right)$$

- Count the number of points, c_1 within B_1 , c_2 within B_2 ...

- For a new test point x

$$p(x) = \sum_{j=1}^n \frac{nc_j}{m} I(x \in B_j)$$



Why is histogram valid?

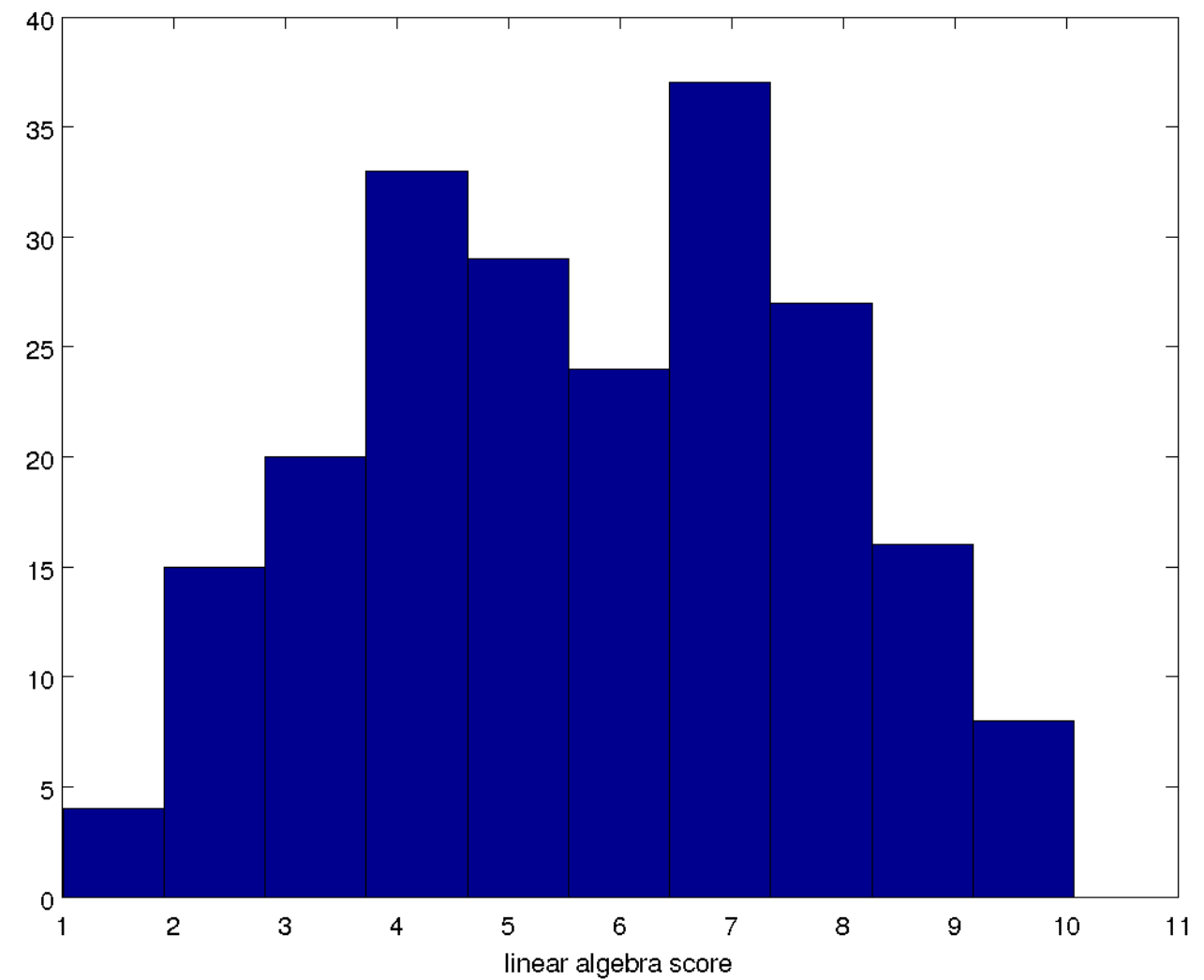
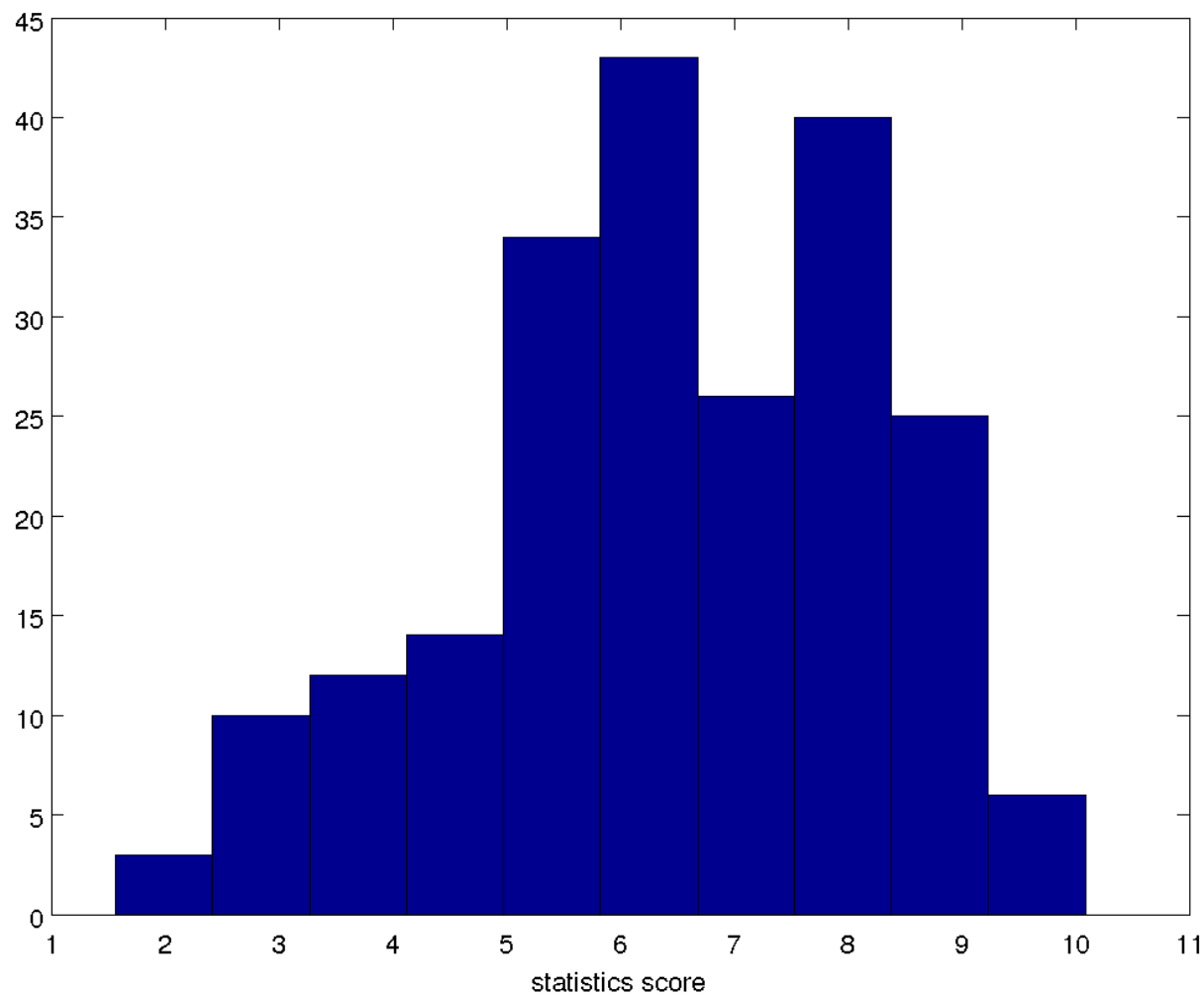
- Requirement for density $p(x)$
- $p(x) \geq 0, \int_{\Omega} p(x)dx = 1$

- For histogram,

$$\begin{aligned}\int_{[0,1)} p(x)dx &= \int_{[0,1)} \sum_{j=1}^n \frac{nc_j}{m} I(x \in B_j) dx \\ &= \sum_{j=1}^n \int_{\left[\frac{j-1}{n}, \frac{j}{n}\right)} \frac{nc_j}{m} dx \\ &= \sum_{j=1}^n \frac{c_j}{m} = 1\end{aligned}$$

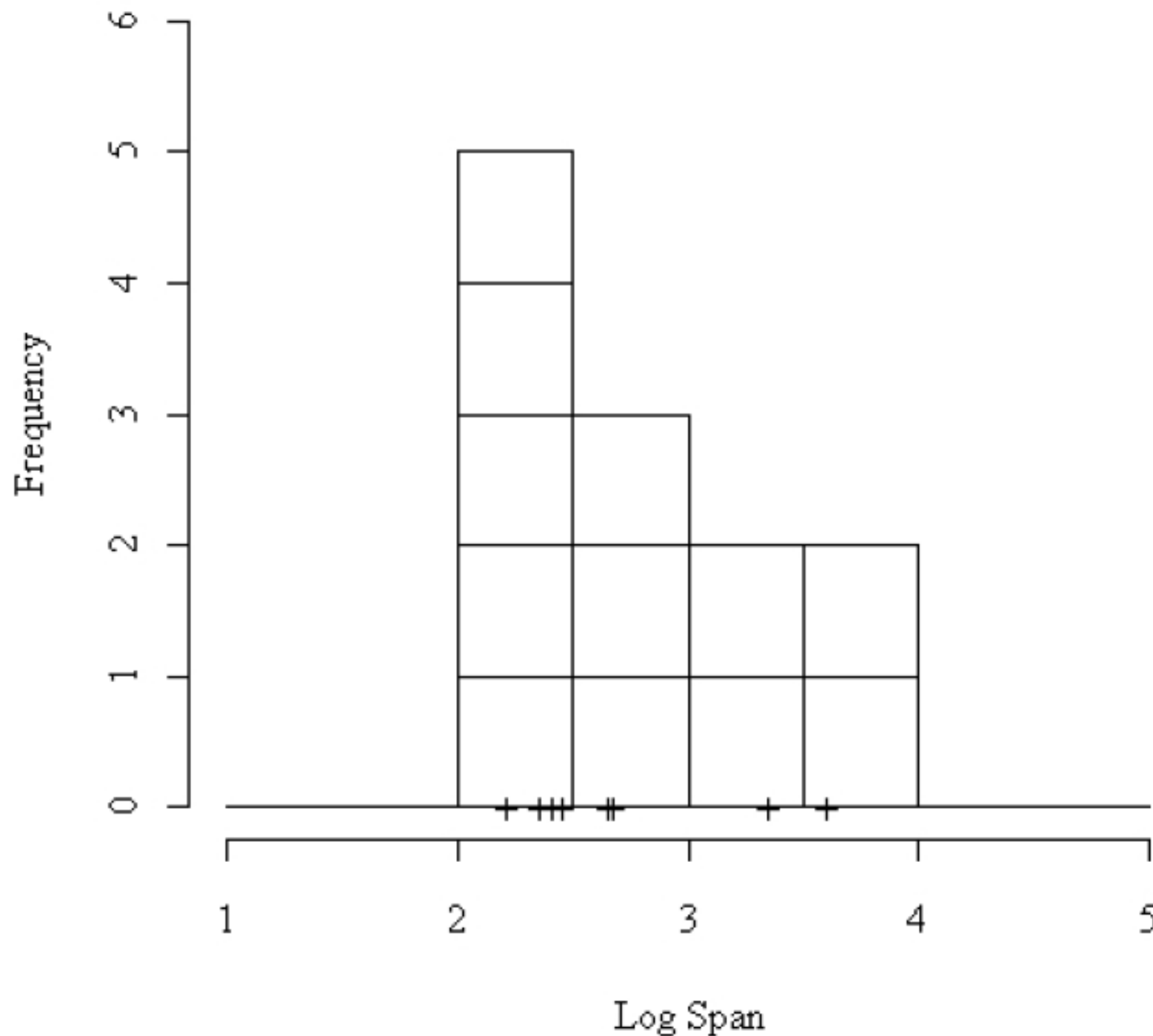
Example: test scores (cont.)

- What is missing if we want density?



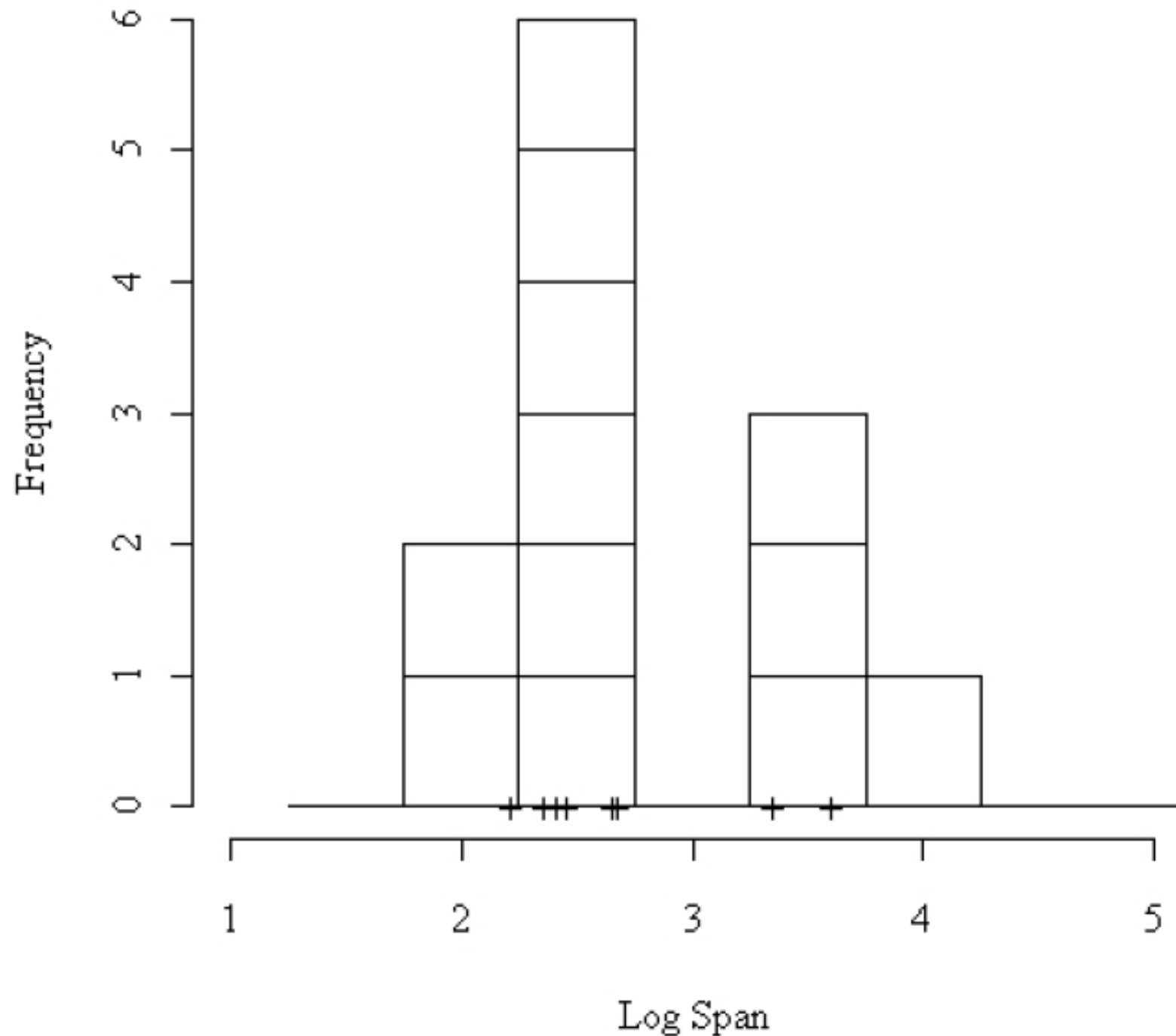
Output depends on where you put the bins

**Histogram with breaks at n.0 and n.5
binwidth=0.5**



Output depends on where you put the bins

**Histogram with breaks at n.25 and n.75
binwidth=0.5**



Higher dimensional histogram

- Given m iid samples $\mathcal{D} = \{x^1, x^2, \dots, x^m\}$, $x^i \in [0,1)^d$

- Split $[0,1)^d$ evenly into n^d bins

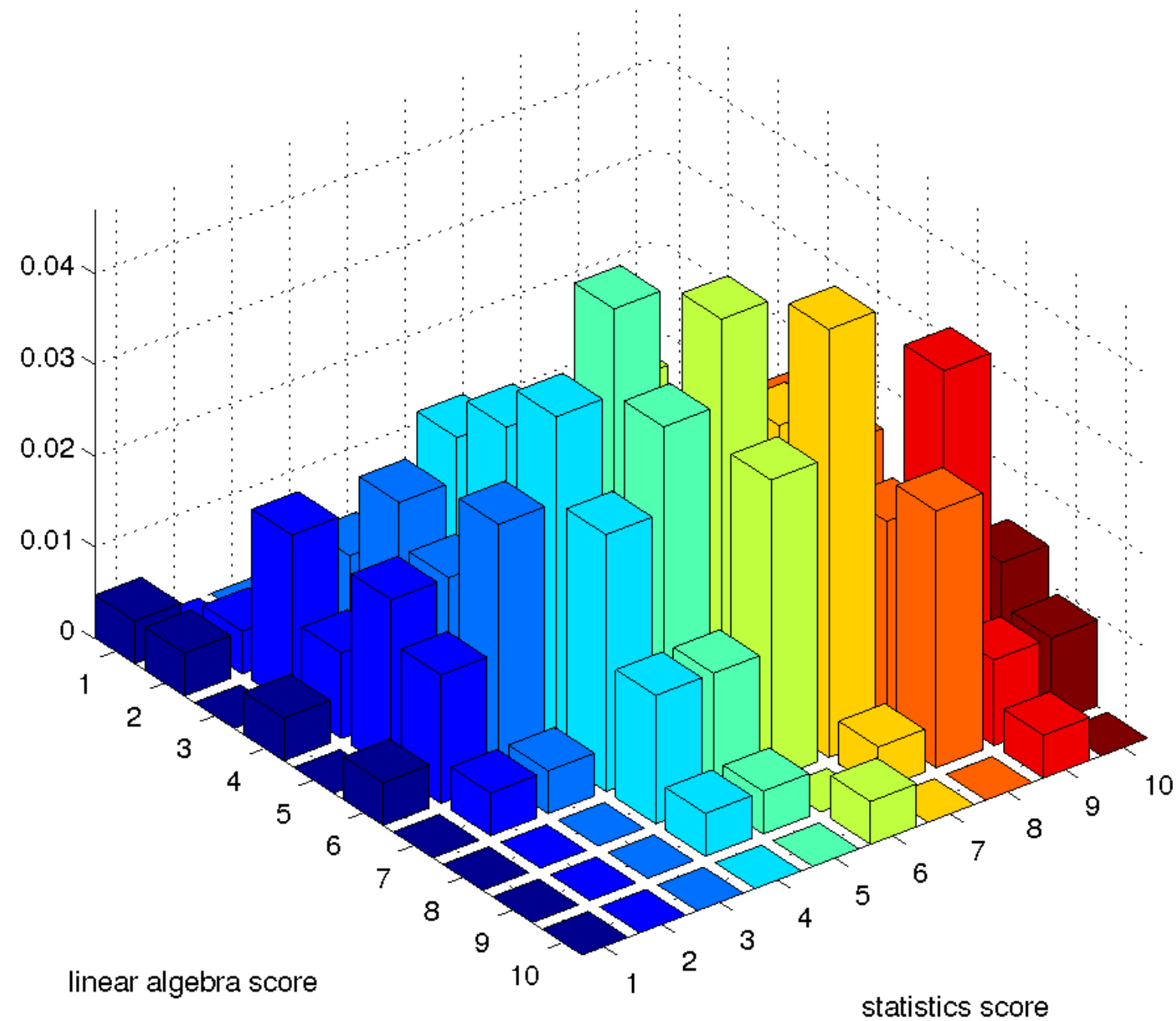
$$B_1 = \left[0, \frac{1}{n}\right) \times \left[0, \frac{1}{n}\right) \dots \times \left[0, \frac{1}{n}\right),$$

$$B_2 = \left[\frac{1}{n}, \frac{2}{n}\right) \times \left[0, \frac{1}{n}\right) \dots \times \left[0, \frac{1}{n}\right),$$

$$\dots$$
$$B_{n^d} = \left[\frac{n-1}{n}, 1\right) \times \left[\frac{n-1}{n}, 1\right) \dots \times \left[\frac{n-1}{n}, 1\right)$$

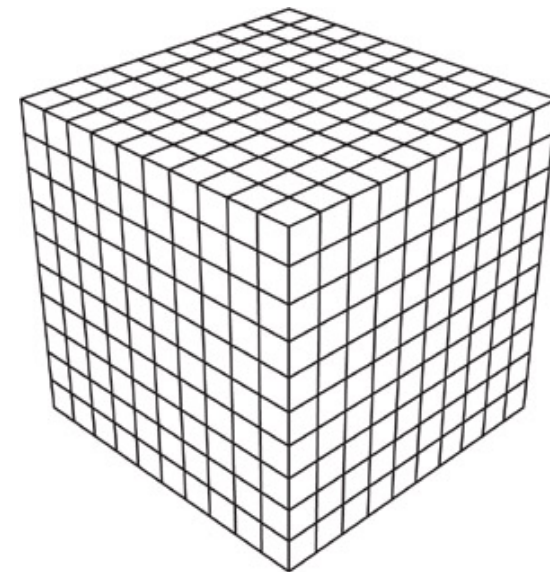
- Bin size is $h = \frac{1}{n}$

Class scores



Computation and statistical considerations

- Problem I: **too many bin! Not good for high dimensional data**
 - If n^d is larger than m , most bins are empty
 - Eg. $n = 10, d = 6$, need ~ 1 million data points



- Problem II: statistically histogram is not the best

- Integrated risk:

$$r(\hat{p}, p) := \int_R \mathbb{E}_X \left[(\hat{p}(x) - p(x))^2 \right] dx$$

- Histogram (with bin size $h \sim m^{-1/3}$)

$$r(\hat{p}, p) \sim \frac{C}{m^{2/3}}$$

- Kernel density estimator (with bandwidth $h \sim m^{-1/5}$)

$$r(\hat{p}, p) \sim \frac{C}{m^{4/5}}$$

- Difference even big for high dimensional data

Kernel density estimation

- Kernel density estimator

$$p(x) = \frac{1}{m} \sum_i^m \frac{1}{h} K\left(\frac{x^i - x}{h}\right)$$

- Smoothing kernel function

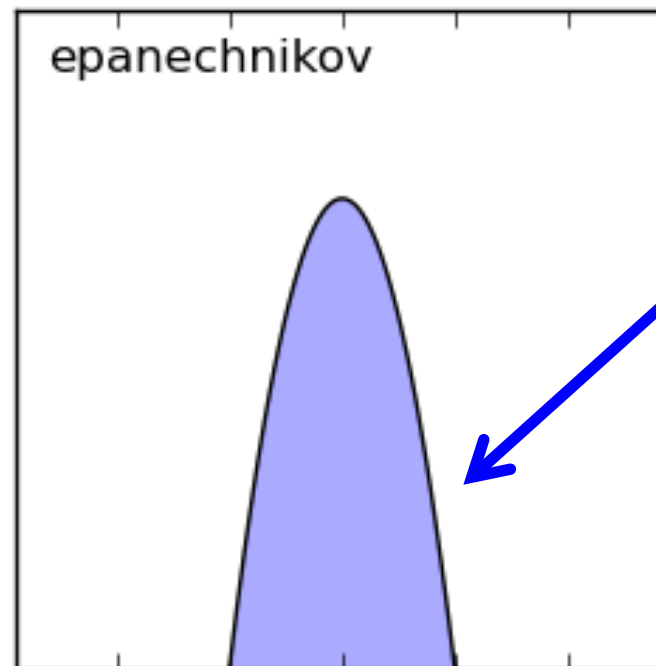
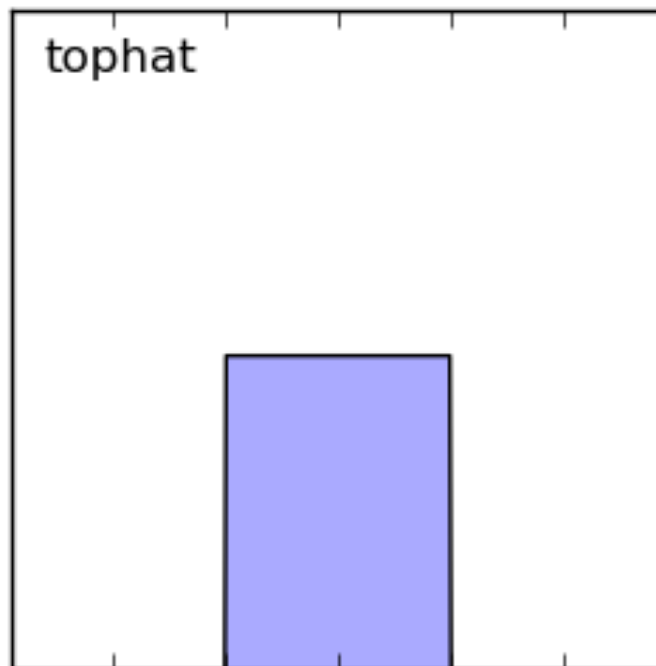
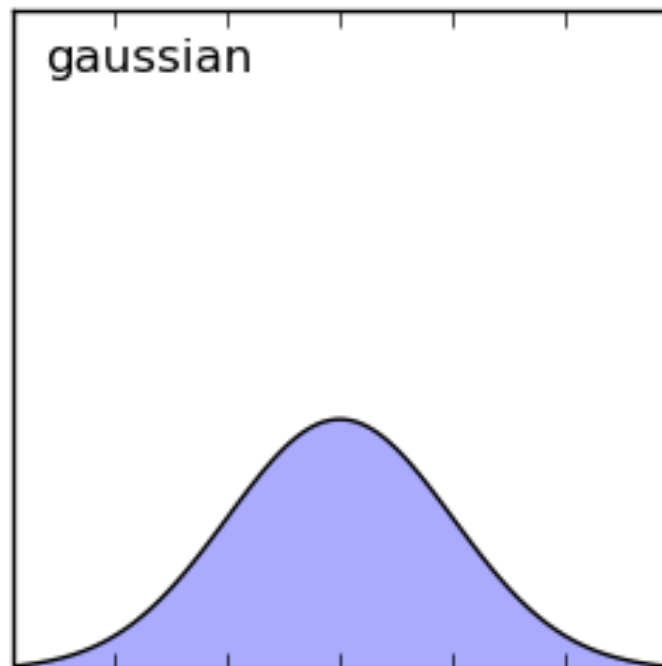
- $K(u) \geq 0$,
- $\int K(u)du = 1$,
- $\int uK(u) = 0$,
- $\int u^2 K(u)du < \infty$

- An example: Gaussian kernel $K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$

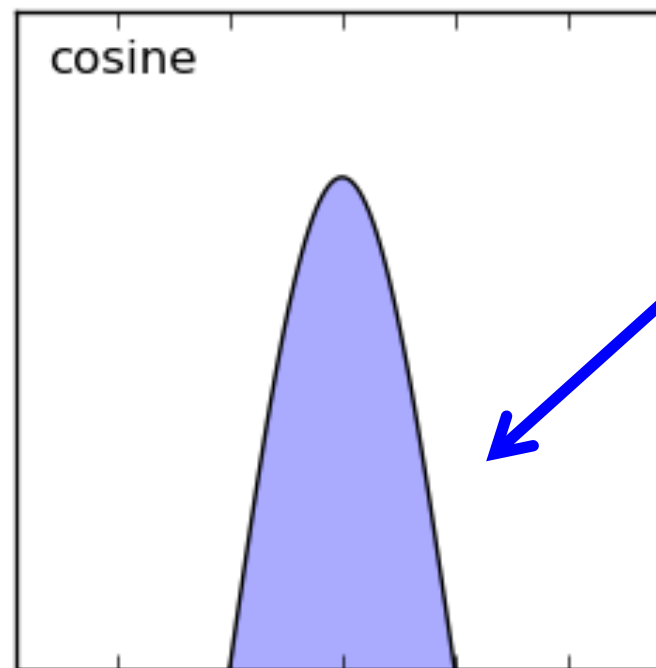
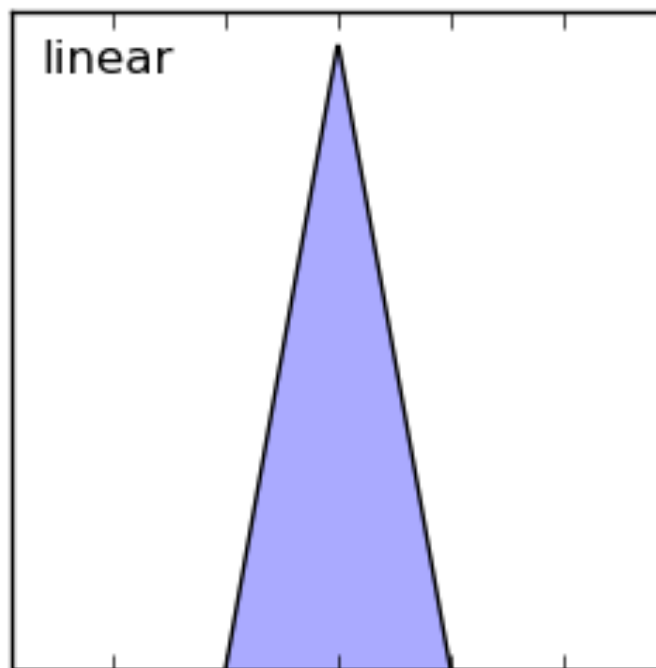
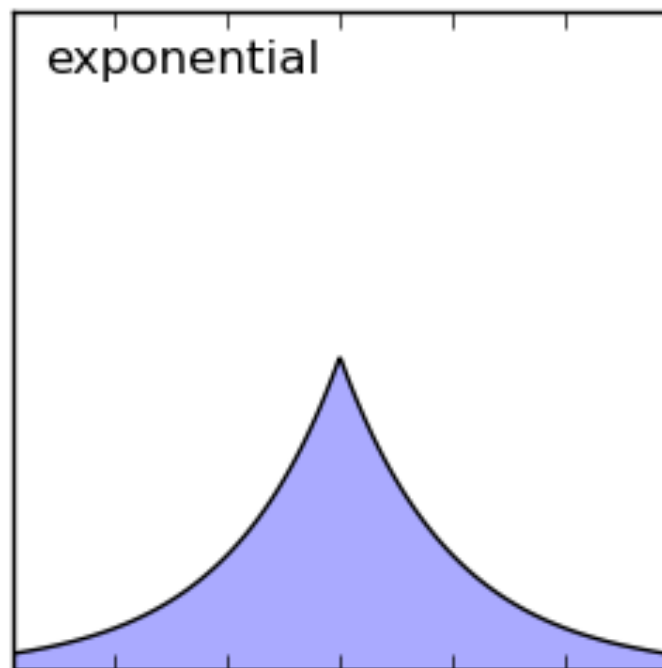
Smoothing kernel functions

- An example: Gaussian kernel $K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$

Available Kernels

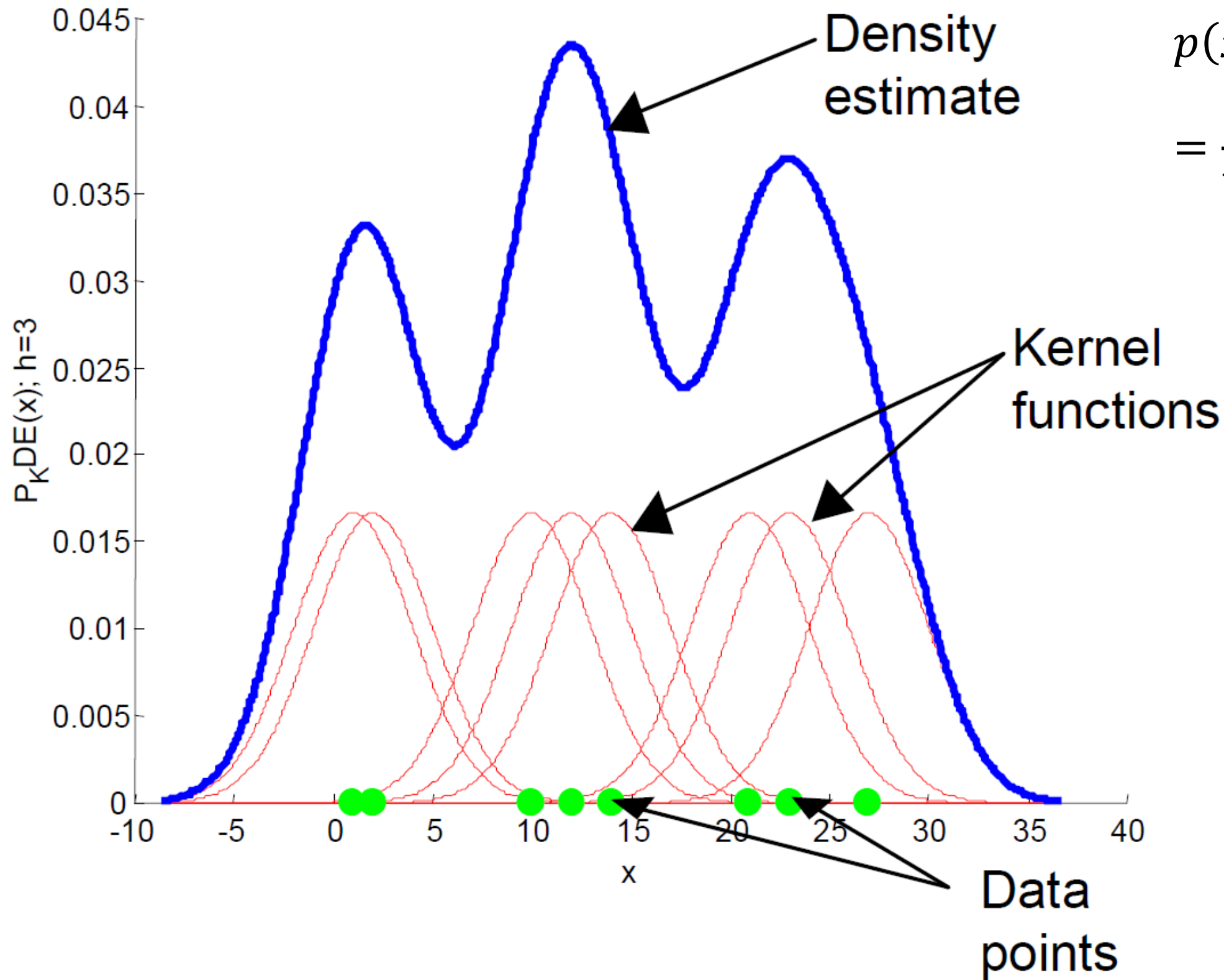


$$K(u) = \frac{3}{4}(1 - u^2)I(|u| \leq 1)$$



$$K(u) = \frac{\pi}{4} \cos\left(\frac{\pi}{2}u\right)I(|u| \leq 1)$$

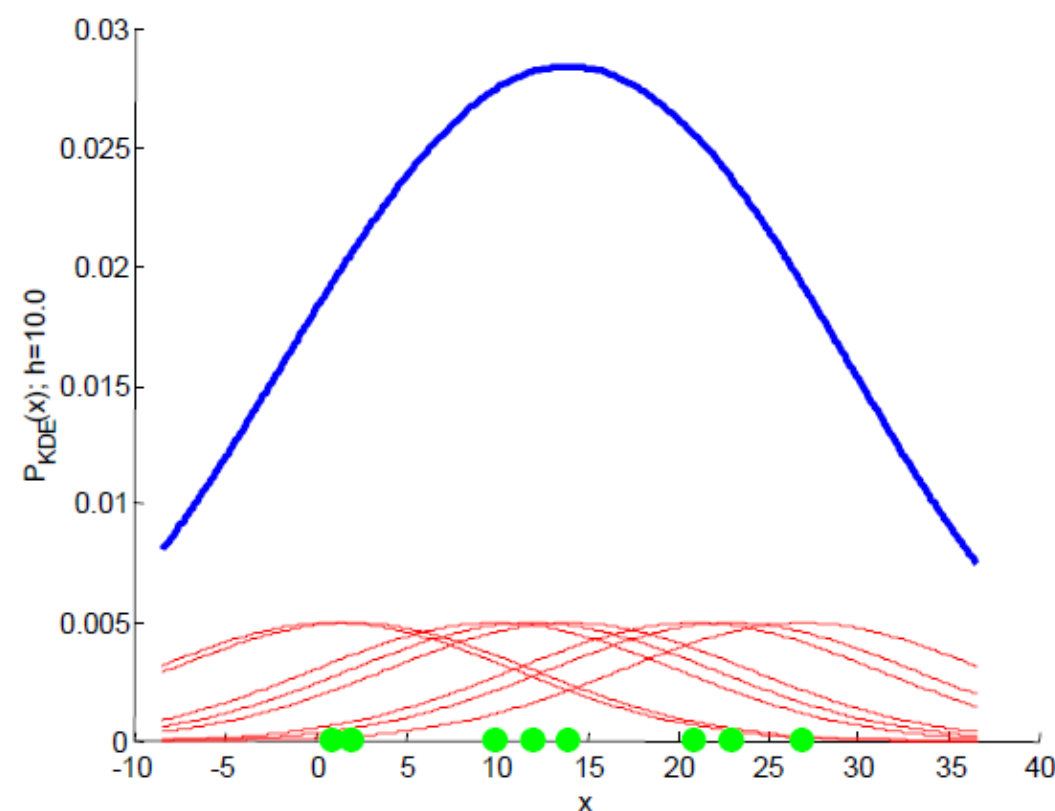
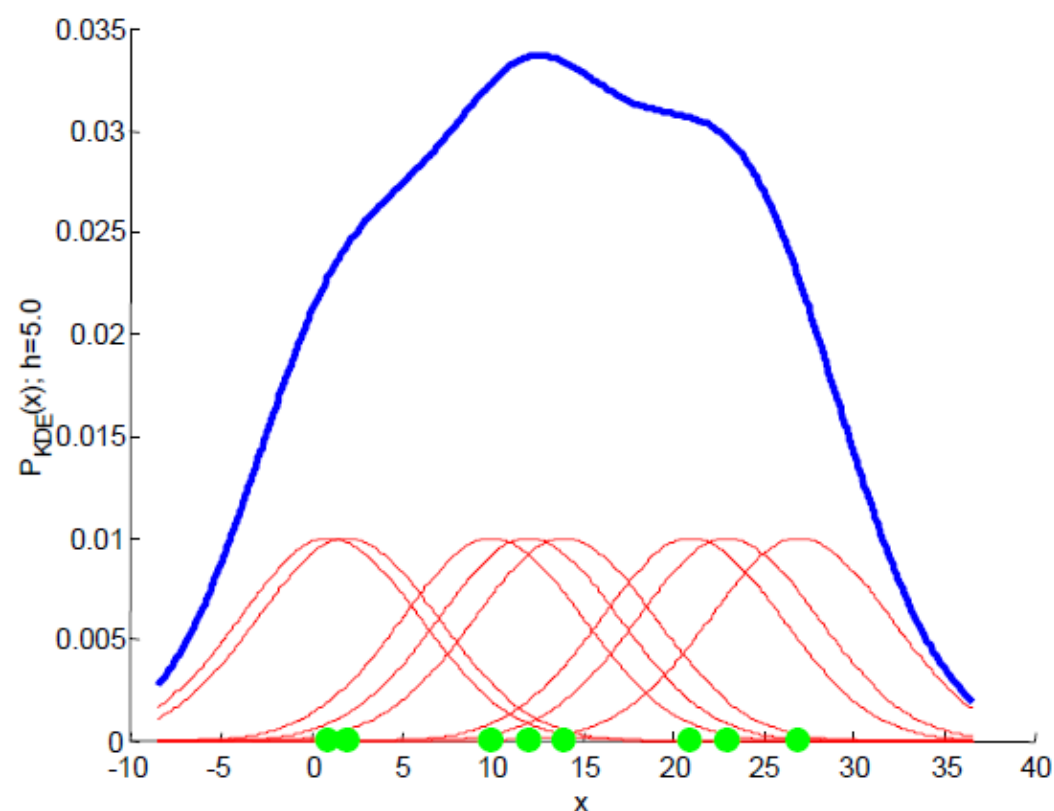
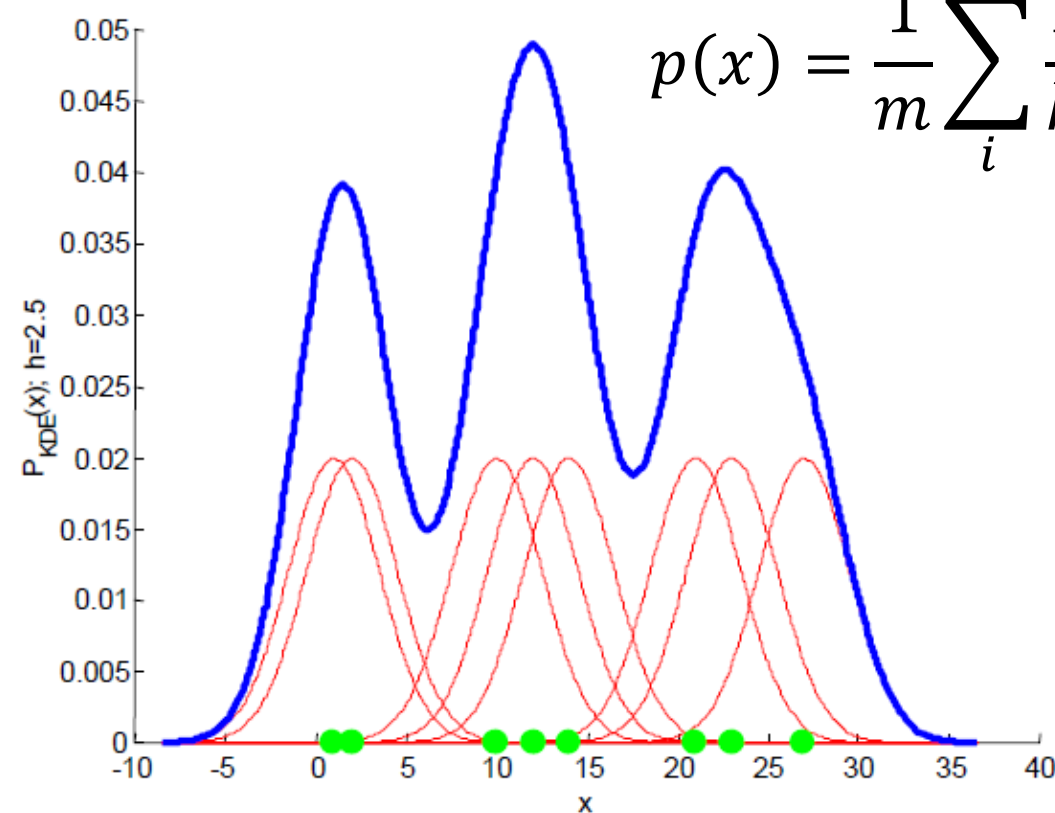
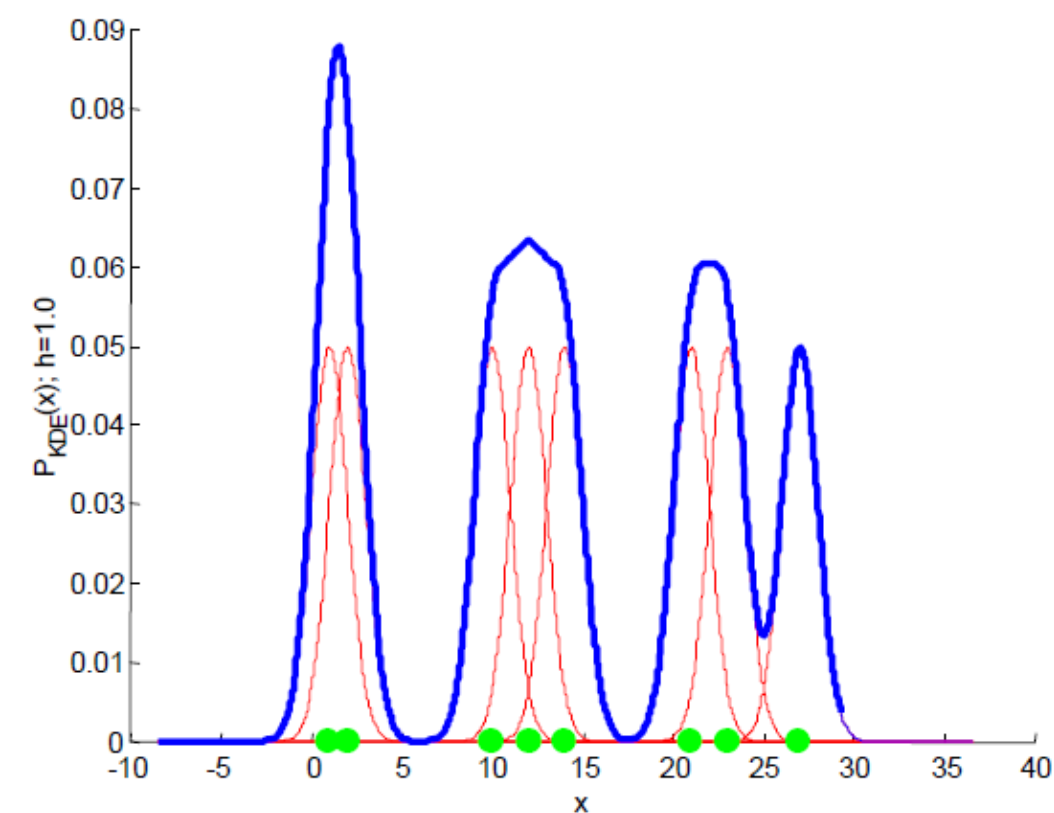
Example



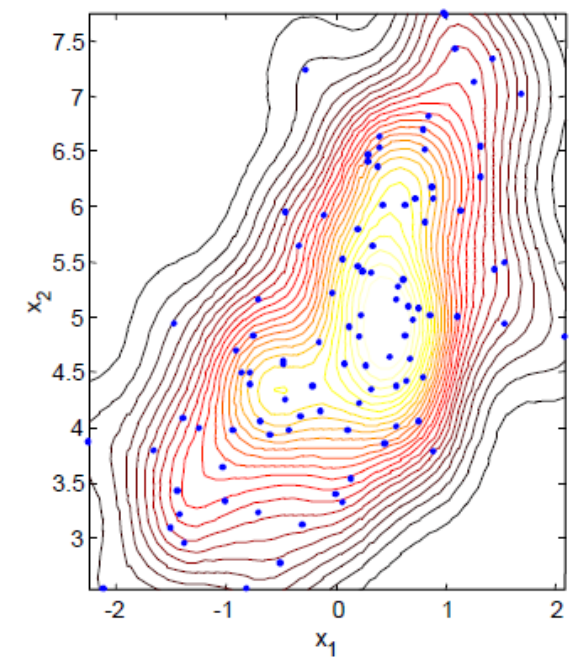
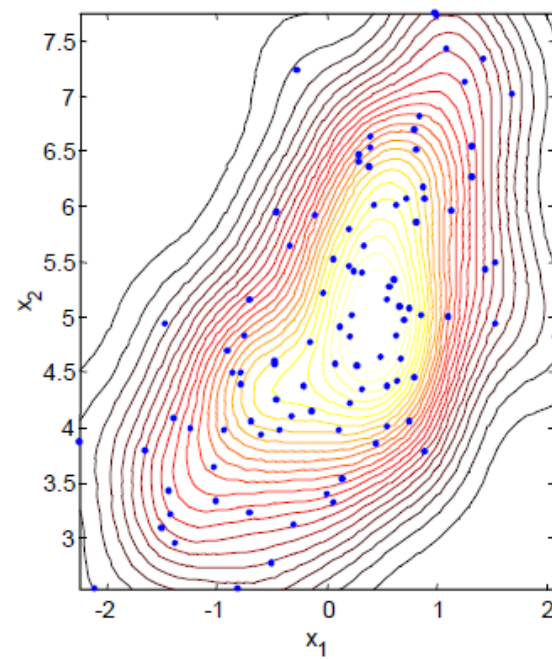
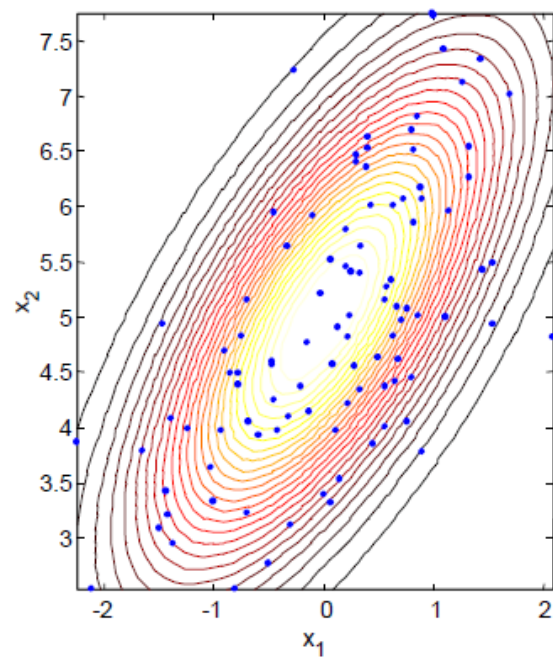
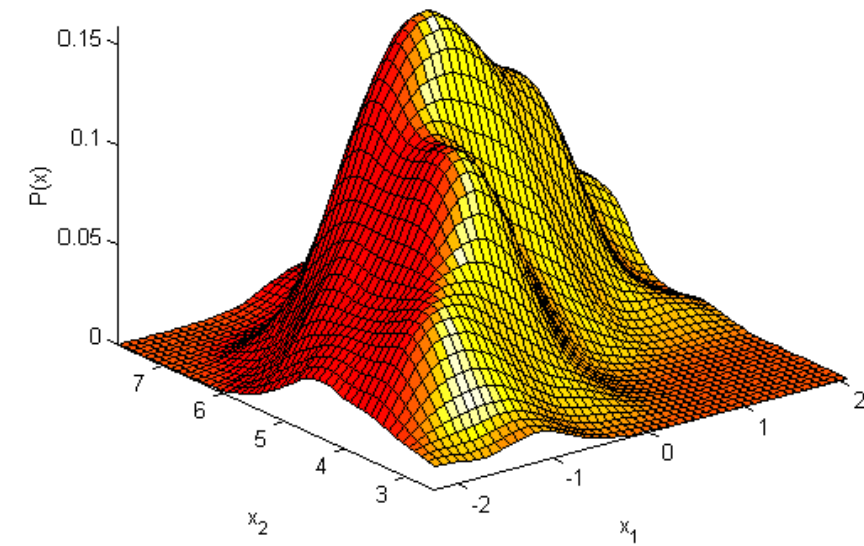
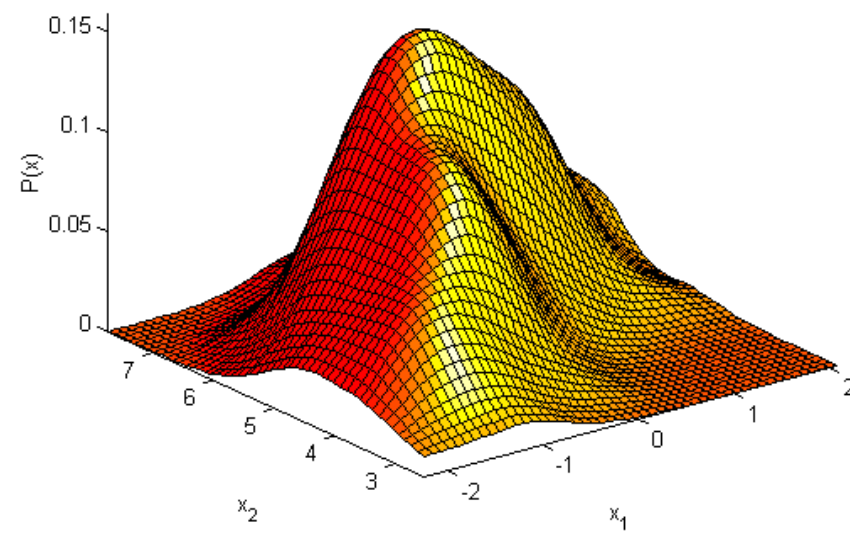
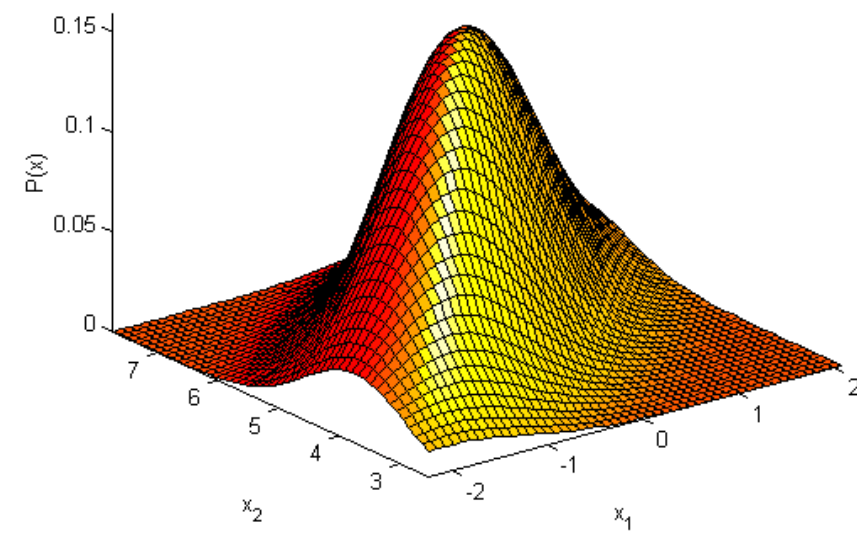
$$p(x) = \frac{1}{m} \sum_i^m \frac{1}{h} K\left(\frac{x^i - x}{h}\right)$$

Effect of the kernel bandwidth h

$$p(x) = \frac{1}{m} \sum_i^m \frac{1}{h} K\left(\frac{x^i - x}{h}\right)$$



Two-dimensional example



What is the best kernel bandwidth?

- Silverman's rule of thumb: If using the Gaussian kernel, a good choice for h is

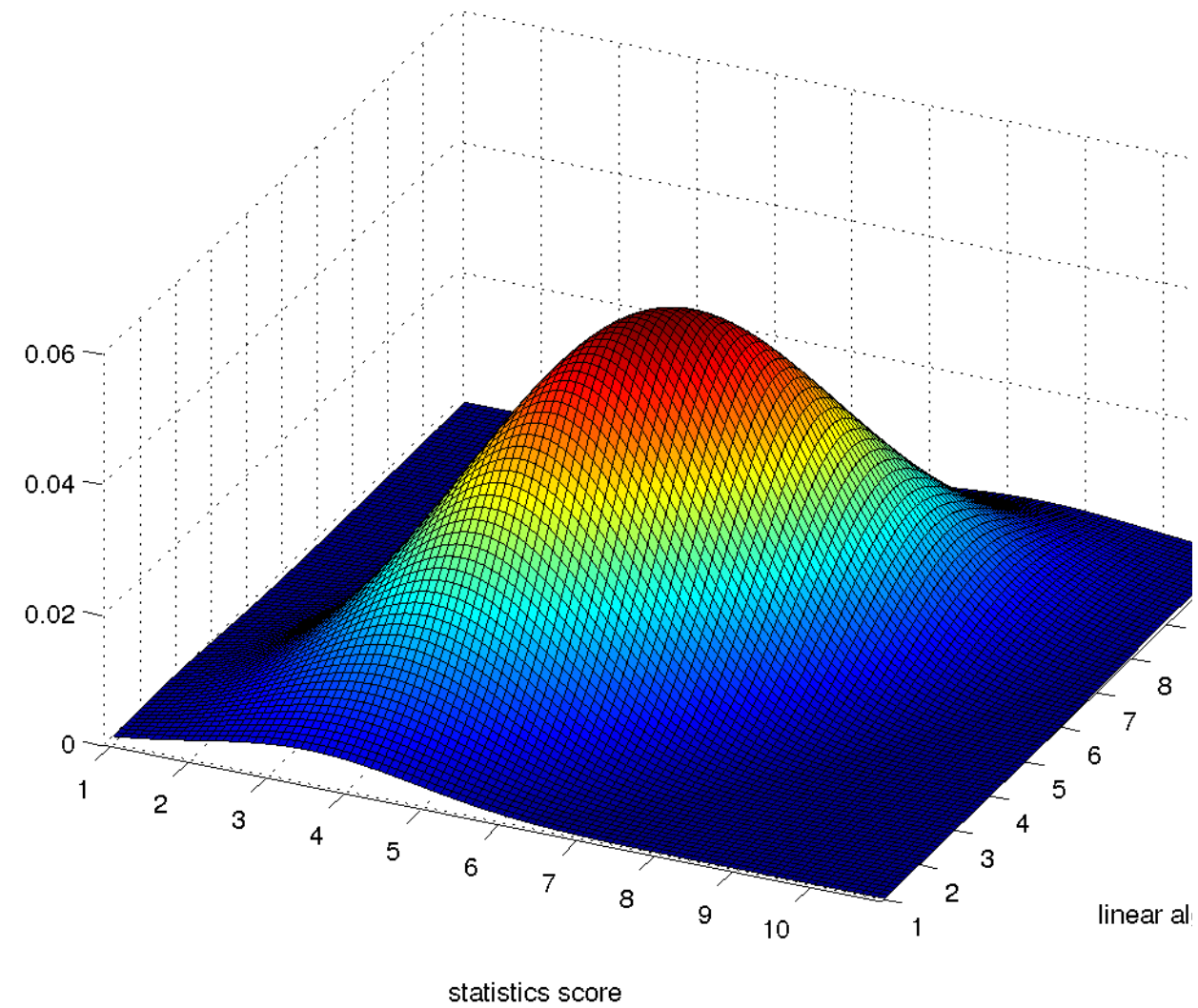
$$h \approx 1.06 \hat{\sigma} m^{-1/5}$$

where $\hat{\sigma}$ is the standard deviation of the samples

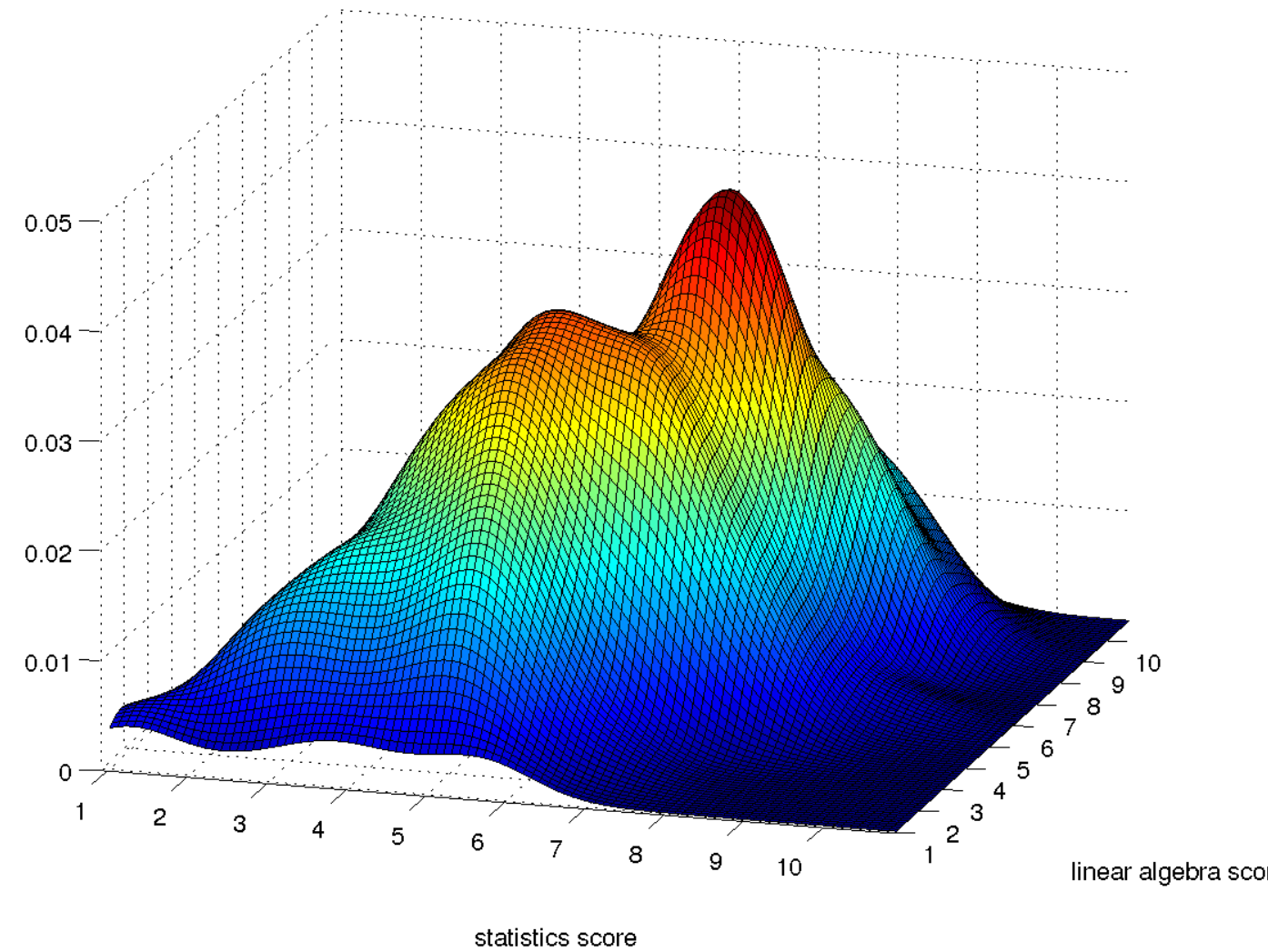
- A better but more computational intensive approach:
 - Randomly split the data into two sets
 - Obtain a kernel density estimate for the first
 - Measure the likelihood of the second set
 - Repeat over many random splits and average

Parametric vs. nonparametric

Gaussian density

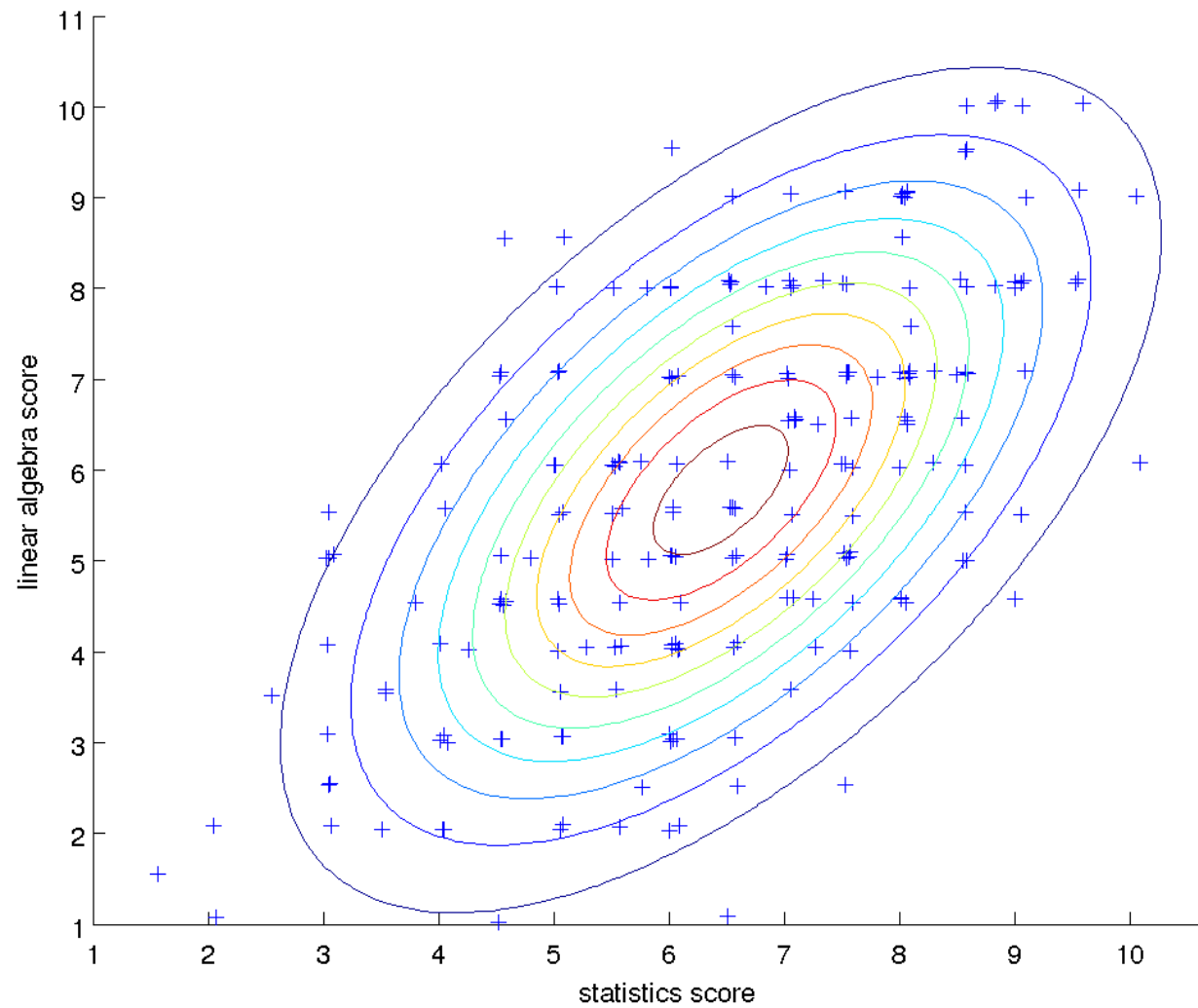


kernel density estimator

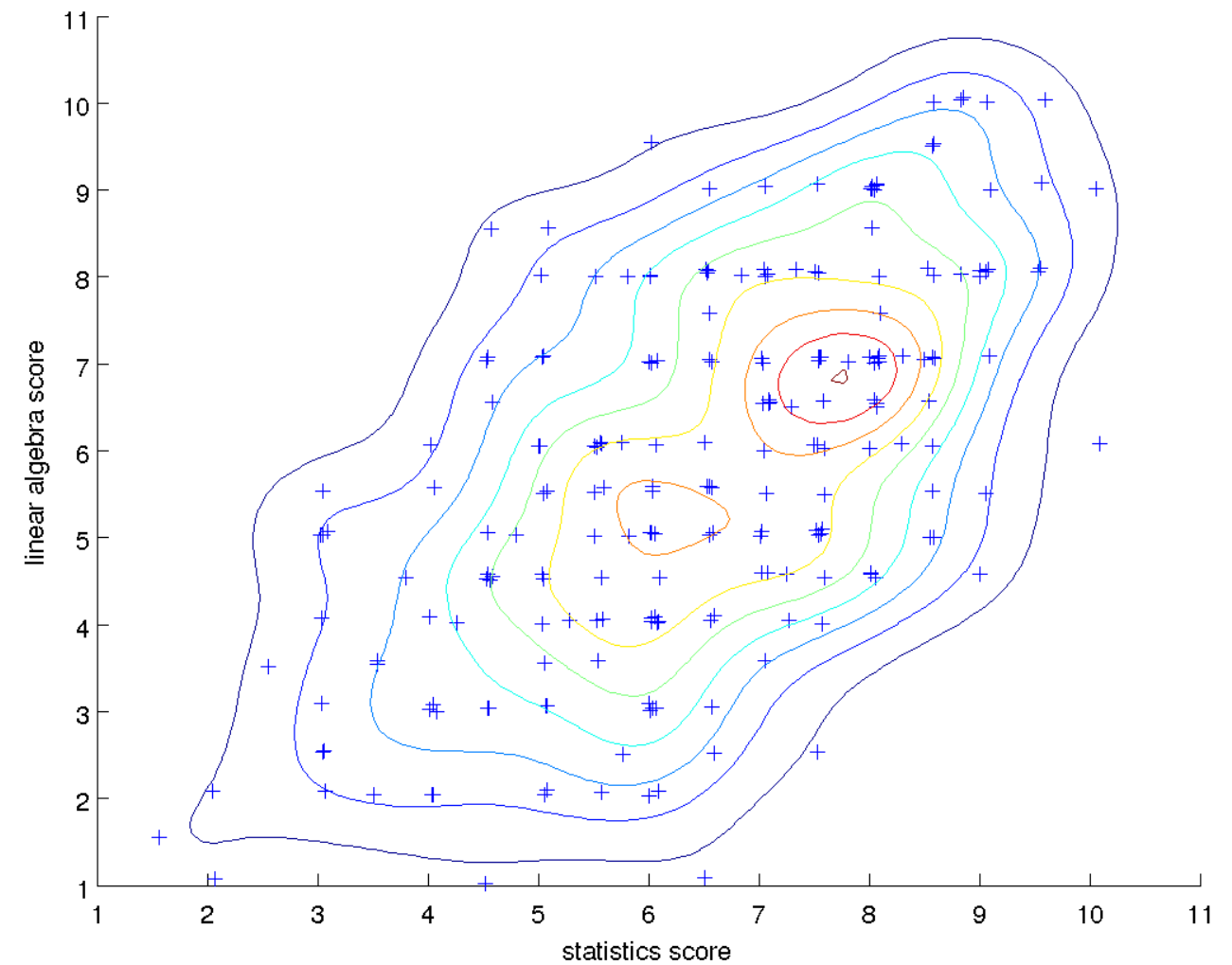


Parametric vs. nonparametric

statistics vs linear algebra (Gaussian)



statistics vs linear algebra



Parametric vs. nonparametric

- Nonparametric models place very mild assumptions on the data distribution and provide good models for complex data

Parametric models rely on very strong (simplistic) distributional assumptions

- Nonparametric models (not histograms) requires storing and computing with the entire data set.

Parametric models, once fitted, are much more efficient in terms of storage and computation.